Market Power and Firm-Level Welfare Impact in Production Networks*

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Abstract

Which firms matter most for the welfare cost of market power? This paper develops firm-level welfare attribution measures for markup distortions in production networks. Using a quadratic welfare representation, we map firm-level markups and firm-to-firm linkages into aggregate welfare losses, marginal welfare effects, and finite single-firm welfare gains. We implement the framework using administrative firm-to-firm transaction data for Belgium from 2002 to 2014. Aggregate Total Welfare Cost roughly doubles over the sample period, rising from 0.0267 log points in 2002 to 0.0528 log points in 2014, and production linkages amplify these losses relative to a horizontal benchmark. Beneath the aggregate loss, welfare impact is highly concentrated: most firms have small effects, while a small upper tail accounts for most positive firm-level welfare impact. High-impact firms are persistent over time, and their rankings remain stable under Ramsey benchmarks that allow markups to finance fixed costs. The results show that aggregate welfare losses mask substantial heterogeneity in the firms that account for welfare-relevant markup distortions.

1 Introduction

Which firms matter most for the welfare cost of market power? The answer is not obvious in a production network. A markup charged by one firm raises costs not only for its direct customers, but also for downstream firms that use those customers' outputs, and for the rest of the economy through input–output reallocation. Market power is therefore measured at the level of firms and markets, but its welfare cost is a general-equilibrium object. The firms that matter most for welfare need not be the firms with the largest sales, the highest markups, or the largest market shares.

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This paper develops firm-level welfare attribution measures for markup distortions in production networks. The motivation is policy-relevant but broader than any single policy instrument: many interventions are local, while the welfare effects of market power are not. Antitrust investigations, merger reviews, regulatory interventions, and market studies target particular firms, markets, or supplier–buyer relationships. Aggregate welfare measures summarize the total cost of distortions, but do not identify where a markup-wedge reduction has the largest payoff. Firm-level measures of market power identify where markups are high, but do not measure how much welfare would change if a particular markup wedge changed. The missing object is a welfare-based ranking of firms that combines markup wedges, economic exposure, network position, and the reallocation induced by a change in markups.

The starting point is the Total Welfare Cost of markups, defined as the log output loss relative to a benchmark allocation. Following Baqaee and Farhi (2020), aggregate welfare losses can be represented as network-generalized Harberger triangles: markup wedges propagate through the production network, and welfare losses depend on the dispersion of propagated wedges within buyers' input bundles. We use this quadratic structure to move from aggregate welfare accounting to firm-level welfare attribution.

The framework delivers two firm-level welfare objects. The first is the marginal welfare effect, denoted by M_k. It is the derivative of aggregate welfare losses with respect to firm k's markup wedge and measures the local welfare effect of a small markup change, holding other wedges fixed and allowing for general-equilibrium reallocation. The second is the finite single-firm welfare gain, denoted by ΔTWC_k. It measures the welfare gain from setting firm k's markup wedge to the benchmark while holding all other wedges fixed. The two objects are linked by the same curvature matrix that governs aggregate welfare losses. This link is useful empirically because the same estimated markups and firm-to-firm expenditure shares discipline the aggregate welfare cost, local firm-level leverage, and discrete single-firm reform gains.

We implement the framework using anonymous administrative firm-to-firm transaction data for Belgium from 2002 to 2014. These data contain the universe of reported domestic firm-to-firm VAT transactions, allowing us to observe the production network directly rather than infer it from sectoral input–output tables (Dhyne, Petrin, et al., 2015; Bernard et al., 2022; Dhyne, Kikkawa, and Magerman, 2022). We combine these transactions with firm balance-sheet data to construct markups, input–output shares, Domar weights, and the welfare objects implied by the model. The empirical setting is well suited to the question because the same data reveal both the wedges and the network through which those wedges propagate.

The empirical analysis delivers three main results. First, aggregate welfare losses from markups are sizable

and rise over time. In the baseline network economy, TWC roughly doubles from 0.0267 log points in 2002 to 0.0528 log points in 2014. A horizontal benchmark that shuts down firm-to-firm input propagation yields lower welfare losses, showing that production linkages amplify the aggregate cost of markup dispersion. Production-network structure therefore changes the measured welfare cost of market power.

Second, welfare impact is highly concentrated, but not captured by firm size alone. Truncation exercises show that large firms account for a substantial share of aggregate TWC. At the same time, replacing omitted firms by a residual node substantially increases measured losses relative to a firm-only truncation, especially when only a small number of top firms is retained. Large firms matter, but their welfare impact cannot be measured accurately by isolating them from the firms around them.

The same concentration appears in the firm-level welfare objects. Both M_k and ΔTWC_k are strongly right-skewed: most firms have small welfare effects, while a small upper tail accounts for a large share of total positive firm-level welfare impact. The welfare-relevant firms are those whose markups interact with high economic exposure and network positions that make their distortions costly for the rest of the economy. Ranking firms by welfare impact therefore reveals information beyond any single primitive such as sales, markups, or market shares.

Third, the identity of high-impact firms is persistent and robust. Firms with high marginal or finite welfare impact tend to remain high-impact over time. Rank-persistence regressions, top-group survival rates, and quintile transition matrices all show substantial stability in the upper tail. Firm-level rankings also remain highly correlated across second-best Ramsey benchmarks that allow part of observed markups to finance fixed costs. Fixed costs affect the level of measured welfare losses more than they affect the identity of the firms with the largest welfare impact.

The paper contributes by turning aggregate welfare accounting for markup distortions into firm-level welfare attribution. Work on market power measures firm-level markups and studies their aggregate consequences (Hall, 1988; De Loecker and Warzynski, 2012; De Loecker, Eeckhout, and Unger, 2020; Edmond, Midrigan, and Xu, 2023). The misallocation literature studies the aggregate costs of wedges across firms and sectors (Harberger, 1954; Hsieh and Klenow, 2009; Restuccia and Rogerson, 2008). The production-network literature shows how shocks and distortions propagate through input–output linkages (Hulten, 1978; Acemoglu et al., 2012; Gabaix, 2011; Baqaee and Farhi, 2019; Baqaee and Farhi, 2020; Bigio and La'O, 2020), and related work studies policy and distortions in network economies (Liu, 2019). We combine these ingredients to construct a mapping from firm-level markups and firm-to-firm linkages to aggregate welfare losses, marginal welfare effects, and finite

single-firm welfare gains. Empirically, the paper shows that welfare impact is concentrated, persistent, and shaped by production-network position, so the firms that matter most for welfare are not identified by markup levels or size alone.

The remainder of the paper proceeds as follows. Section 2 presents the theoretical framework. Section 3 describes the data. Section 4 reports the empirical results. Section 5 concludes.

2 Theoretical Framework

This section develops the welfare accounting framework used in the empirical analysis. The goal is to map firm-level markup wedges and firm-to-firm input linkages into three objects: the aggregate Total Welfare Cost of markups, the marginal welfare effect of each firm's markup wedge, and the finite single-firm welfare gain from setting one firm's wedge to the benchmark.

The framework builds on the general-equilibrium approach of Baqaee and Farhi (2020), which extends classical Harberger-type welfare accounting to economies with input–output linkages and endogenous reallocation. Markup wedges propagate through the production network, and welfare losses arise from dispersion in propagated wedges across buyers' input bundles. The same quadratic object that measures aggregate welfare losses also delivers the firm-level welfare objects used below.

We take technologies, preferences, and the production network as primitives and evaluate welfare conditional on the observed distorted equilibrium. Markups create wedges between prices and marginal costs, generating allocative distortions that propagate through supply chains. The framework maps these wedges into welfare objects using revenue shares, substitution elasticities, and network exposure.

2.1 Economic Environment and Network Objects

We consider a closed general-equilibrium economy populated by a set of producing firms indexed by $i = 1, \ldots, N$, a representative household that consumes final goods and owns all firms, and a single primary factor that aggregates labor and capital into value added. Firms produce differentiated goods that may be sold either to other firms as intermediate inputs or to the household for final consumption.

Final demand. The representative household aggregates final goods through a CES aggregator,

$$C = \left(\sum_{i=1}^{N} \omega_{0i}^{1/\theta_0} c_i^{(\theta_0-1)/\theta_0} \right)^{\theta_0/(\theta_0-1)}, \qquad \sum_{i=1}^{N} \omega_{0i} = 1, \tag{1}$$

where $\theta_0 > 0$ is the elasticity of substitution across final goods. Let $b_i \equiv p_i c_i / \sum_j p_j c_j$ denote household expenditure shares.

Production and markups. Each firm combines value added and intermediate inputs using a nested CES technology. Let $\theta_j > 0$ denote the elasticity of substitution across suppliers in buyer j's input bundle. Firms minimize costs taking input prices as given. On the output side, firm i charges a constant gross markup $\mu_i \geq 1$ over marginal cost,

$$p_i = \mu_i mc_i. \tag{2}$$

Markups introduce wedges between prices and marginal costs and constitute the source of allocative distortions studied in the paper.

Revenue-based network representation. Let $x_{ij} = p_j m_{ij}$ denote nominal purchases by buyer i from seller j, and let $S_i = p_i y_i$ denote firm i's nominal sales. Define the revenue-based input–output matrix

$$\Omega_{ij} = \frac{x_{ij}}{S_i}, \qquad i, j \in \{1, \ldots, N, \text{VA}\}, \tag{3}$$

where rows index buyers and columns index sellers. The associated revenue-based Leontief inverse is

$$\Psi \equiv (I - \Omega)^{-1}. \tag{4}$$

Revenue-based Domar weights are defined as

$$\lambda_i \equiv \frac{S_i}{\sum_{j=1}^{N} p_j c_j}, \tag{5}$$

and satisfy the identity

$$\lambda' = b' \Psi. \tag{6}$$

Cost-based objects. The transaction data measure expenditure shares at distorted prices. The welfare formulas, however, require cost shares evaluated at marginal cost. We therefore define the cost-based input–output matrix

$$\widetilde{\Omega} \equiv \operatorname{diag}(\mu)\,\Omega,$$

the associated cost-based Leontief inverse

$$\widetilde{\Psi} \equiv (I - \widetilde{\Omega})^{-1},$$

and cost-based Domar weights

$$\widetilde{\lambda}' \equiv b'\widetilde{\Psi}.$$

Multiplying revenue shares by markups converts expenditure shares measured at distorted prices into cost shares evaluated at marginal cost. As shown by Baqaee and Farhi (2020), this accounting transformation preserves the underlying production network while expressing it in welfare-relevant units.

2.2 Total Welfare Cost of Markups

We now characterize the aggregate welfare loss generated by markup distortions. Let $Y(\mu)$ denote equilibrium real output under the observed markup vector μ, and let Y^* denote the efficient allocation obtained under marginal-cost pricing, holding technologies and the production network fixed. The Total Welfare Cost (TWC) of markups is defined as

$$\mathscr{L}(\mu) \equiv \log\left(\frac{Y^*}{Y(\mu)}\right). \tag{7}$$

This object measures the allocative inefficiency induced by pricing wedges, conditional on the existing production structure. Throughout, welfare is evaluated relative to the competitive benchmark while holding primitives fixed.

Quadratic representation. Let $\Delta \log \mu$ denote the vector of log-markup wedges relative to the competitive benchmark. Define the vector of effective wedges

$$t \equiv \Psi \Delta \log \mu, \tag{8}$$

which incorporates both direct and indirect propagation through the production network.

The formulas below are written in terms of the revenue-based expenditure shares observed in the transaction

data, together with the markup wedges that convert those shares into welfare-relevant cost wedges. Equivalently, the cost-based objects defined above provide the accounting interpretation of the same distorted equilibrium.

Following Baqaee and Farhi (2020), a second-order approximation around the efficient allocation yields

$$\mathscr{L}(\mu) \approx \frac{1}{2} \sum_j \lambda_j \, \theta_j \, \mathrm{Var}_{\Omega(j)}(t)\,, \tag{9}$$

where $\Omega(j)$ denotes the jth row of the revenue-based input–output matrix, and the weighted variance operator is defined as

$$\mathrm{Var}_{\Omega(j)}(x) = \sum_i \Omega_{ji} \big(x_i - \bar{x}_j\big)^2, \qquad \bar{x}_j = \sum_i \Omega_{ji} x_i.$$

Welfare losses therefore arise from dispersion in effective wedges within each buyer's input bundle. The Domar weight λ_j scales each buyer's contribution by its macroeconomic importance, while θ_j governs the strength of substitution across suppliers.

Economic interpretation. Equation (9) generalizes the classical Harberger triangle to a production network. For each buyer j, welfare losses are proportional to the dispersion of effective input wedges across its suppliers. The effective wedge vector $t = \Psi \Delta \log \mu$ reflects network propagation: a markup charged by an upstream firm raises the effective cost faced by all downstream producers that use its output, directly or indirectly.

Welfare losses are therefore not generated by markup dispersion alone. They are generated by dispersion in propagated wedges within buyers' input bundles. A firm's welfare relevance depends both on its own wedge and on where that wedge enters the production network.

Matrix representation. The quadratic structure can be expressed compactly in matrix form. Let $D_j = \mathrm{diag}(\Omega_{j\cdot})$ and define

$$P_j = D_j - \frac{\Omega_{j\cdot}^\top \Omega_{j\cdot}}{\sum_i \Omega_{ji}},$$

so that $\mathrm{Var}_{\Omega(j)}(t) = t^\top P_j t$. Define

$$H \equiv \sum_j \lambda_j \theta_j P_j.$$

Then welfare losses admit the quadratic form

$$\mathscr{L}(\mu) \approx \frac{1}{2} t^\top H t = \frac{1}{2} (\Delta \log \mu)^\top \underbrace{\big(\Psi^\top H \Psi\big)}_{G} \Delta \log \mu. \tag{10}$$

The matrix H aggregates buyer-level Harberger components, while $G = \Psi^\top H \Psi$ maps seller-level markup wedges into buyer-level dispersion through the production network. The matrix G is symmetric and positive semi-definite and is evaluated at the observed distorted equilibrium under standard regularity conditions. This matrix is the bridge from aggregate welfare accounting to firm-level welfare measurement: its diagonal and off-diagonal elements determine how a change in one firm's markup wedge affects aggregate welfare losses, accounting for its covariance with all other wedges through the production network.

2.3 Firm-Level Marginal Welfare Effects

The aggregate Total Welfare Cost measures the level of misallocation generated by the full vector of markup wedges. We now define a firm-level object that measures the local welfare effect of changing one firm's markup wedge, holding all other markup wedges fixed.

Let $\Delta \log \mu$ denote the vector of markup wedges relative to the relevant benchmark. In the baseline specification, the benchmark is marginal-cost pricing, so $\Delta \log \mu_k = \log \mu_k$. In the second-best specifications below, the benchmark is a Ramsey reference markup vector, so $\Delta \log \mu_k = \log \mu_k - \log \mu_k^\star$.

The marginal welfare effect of firm k is defined as

$$M_k \equiv \frac{\partial \mathscr{L}}{\partial \Delta \log \mu_k}. \tag{11}$$

Using the quadratic representation in equation (10), differentiation yields

$$M_k = \left(G \Delta \log \mu\right)_k, \tag{12}$$

where $G = \Psi^\top H \Psi$ is the curvature matrix of the welfare loss.

Thus, M_k is the kth component of the gradient of aggregate welfare losses with respect to firm-level markup wedges. It measures local welfare leverage: for a small change $d\Delta \log \mu_k$ in firm k's markup wedge,

$$d\mathscr{L} \approx M_k \, d\Delta \log \mu_k. \tag{13}$$

A positive value of M_k means that a marginal increase in firm k's markup raises welfare losses, while a marginal reduction lowers them. Conversely, a negative value reflects second-best interactions: conditional on the existing configuration of distortions, a marginal reduction in one wedge can increase measured welfare losses.

Because $\mathscr{L}(\mu) = \log(Y^*/Y(\mu))$ and the benchmark output Y^* is fixed with respect to the marginal perturbation, the same object can be written as an output elasticity:

$$M_k = -\frac{\partial \log Y}{\partial \log \mu_k}. \tag{14}$$

Computing M_k therefore amounts to computing the general-equilibrium response of aggregate output to a marginal change in firm k's markup. The gradient formula in equation (12) is the definition used for welfare accounting and for the empirical rankings. The decomposition below is not a separate definition of M_k; it is an equivalent one-factor representation used to interpret the same derivative in terms of exposure and reallocation.

The output response has two components. First, there is a direct exposure effect: increasing firm k's markup creates a wedge at a node to which the economy is exposed. Second, there is a reallocation effect: the markup change alters relative prices, induces substitution across suppliers, and changes factor prices. In a one-factor economy, this decomposition can be written as

$$\frac{\partial \log Y}{\partial \log \mu_k} = -\widetilde{\lambda}_k - s_k, \tag{15}$$

where $\widetilde{\lambda}_k$ is the cost-based Domar weight of firm k and s_k summarizes the endogenous factor-price response. Hence,

$$M_k = \widetilde{\lambda}_k + s_k. \tag{16}$$

Let $\Psi_{\cdot k}$ denote column k of the revenue-based Leontief inverse and let $\Psi_{\cdot L}$ denote the total labor-content vector. Define

$$A = \sum_j \lambda_j(\theta_j - 1)\,\mathrm{Var}_{\Omega(j)}\left(\Psi_{\cdot L}\right), \tag{17}$$

$$B_k = \sum_j \lambda_j(\theta_j - 1)\,\mathrm{Cov}_{\Omega(j)}\left(\Psi_{\cdot L}, \Psi_{\cdot k}\right). \tag{18}$$

The wage-normalization term satisfies

$$s_k = \frac{-\lambda_k \Psi_{kL} - B_k}{1 + A}. \tag{19}$$

Substituting into equation (16) gives the implementable expression

$$M_k = \widetilde{\lambda}_k + \frac{-\lambda_k \Psi_{kL} - B_k}{1 + A}. \tag{20}$$

Equation (20) shows that a firm's marginal welfare effect depends not only on its own markup or size, but also on its position in the production network and on the reallocation induced by a markup change. This is why rankings based on M_k need not coincide with rankings based on sales, markups, or Domar weights alone.

Detailed derivations are provided in Appendix D.

2.4 Finite Single-Firm Reforms

The marginal welfare effect M_k characterized in Section 2.3 is a local object. Policy interventions, however, are typically discrete rather than infinitesimal. We therefore characterize the welfare impact of finite firm-level reforms.

Quadratic structure. Welfare losses admit the quadratic representation

$$\mathscr{L}(\Delta \log \mu) = \frac{1}{2}(\Delta \log \mu)^\top G \Delta \log \mu, \tag{21}$$

where G is the symmetric curvature matrix defined in equation (10).

Consider a reform that changes firm k's markup wedge from $\Delta \log \mu_k$ to $\Delta \log \mu_k'$, holding all other wedges fixed. Let $\Delta \log \mu'$ denote the post-reform wedge vector. The associated welfare gain is

$$\Delta TWC_k = \mathscr{L}(\Delta \log \mu) - \mathscr{L}(\Delta \log \mu'). \tag{22}$$

Single-firm-to-benchmark reform. A particularly relevant counterfactual sets firm k's markup wedge to zero, leaving all other wedges unchanged. In this case, the quadratic structure implies

$$\Delta TWC_k = \Delta \log \mu_k \cdot M_k - \frac{1}{2}(\Delta \log \mu_k)^2 G_{kk}, \tag{23}$$

where $M_k = (G \Delta \log \mu)_k$ and G_{kk} denotes the kth diagonal element of G.

Interpretation. Finite welfare gains depend on both the marginal effect and the curvature of the welfare function. The first term in equation (23) corresponds to the integrated marginal effect evaluated at the pre-reform allocation. The second term reflects the nonlinearity of welfare losses in markup wedges and captures how curvature moderates the total gain from eliminating a distortion.

Thus, while M_k measures local welfare leverage, G_{kk} determines how rapidly that leverage declines as the wedge is reduced. Finite reforms are therefore governed entirely by the same curvature matrix G that characterizes aggregate welfare losses and marginal effects. In the empirical analysis, this object is reported as the finite single-firm welfare gain, ΔTWC_k.

2.5 Horizontal Benchmark

The preceding objects are computed in the baseline production network. We also use benchmark economies to clarify which features of the baseline results are driven by network propagation and which depend on the choice of welfare benchmark. The first benchmark isolates the role of production-network linkages by computing aggregate welfare losses in a horizontal economy. This benchmark shuts down firm-to-firm intermediate-input propagation while keeping the distribution of firm-level markups and final-demand weights fixed.

Formally, the horizontal economy imposes

$$\Omega_{FF} = 0,$$

so that firms do not use intermediate inputs from one another. The associated firm-to-firm Leontief inverse is therefore the identity matrix,

$$\Psi_{FF} = I,$$

and effective wedges coincide with firm-level markup wedges:

$$t = \Delta \log \mu .$$

In this benchmark, welfare losses arise from dispersion in firm-level markup wedges across final demand rather than from dispersion in propagated wedges within buyers' input bundles. Let λ_k^H denote the value-added share of firm k in the horizontal economy. The aggregate horizontal Total Welfare Cost is

$$\mathscr{L}^H = \frac{1}{2}\theta_0 \operatorname{Var}_{\lambda^H}(\Delta \log \mu), \tag{24}$$

where

$$\operatorname{Var}_{\lambda^H}(x) = \sum_k \lambda_k^H \left(x_k - \sum_i \lambda_i^H x_i \right)^2 .$$

The horizontal benchmark is useful because it separates two sources of welfare losses. The first is the dispersion of firm-level markup wedges itself. The second is the amplification of these wedges through input–output linkages. Comparing $\mathscr{L}^H$ to the baseline network measure $\mathscr{L}$ therefore quantifies how much production networks amplify the aggregate welfare cost of markups.

This benchmark changes the economic structure by eliminating firm-to-firm intermediate-input use. It should therefore be interpreted as a counterfactual economy without production-network propagation, not as a mechanical truncation of the Leontief inverse in the observed economy.

2.6 Fixed Costs and Second-Best Benchmarks

The second set of benchmarks asks whether the firm-level conclusions depend on treating all observed markup wedges as distortions relative to marginal-cost pricing. The baseline welfare measures evaluate observed markups relative to the first-best allocation. However, if firms face fixed operating costs, marginal-cost pricing may not be feasible. In that case, some positive markup may be required to finance fixed costs, and the relevant benchmark is second-best rather than first-best.

This distinction affects the interpretation of markups, but not the network logic of the welfare formulas. Following McKenzie (1959) and Baqaee and Farhi (2020), fixed costs can be represented through fixed factors supplied in fixed quantity. Under this representation, payments to fixed factors absorb non-convex costs, while marginal substitution conditions across variable inputs remain unchanged. Therefore, the production-network propagation structure, the Leontief inverse, and the curvature objects used above continue to characterize the welfare cost of markup wedges.

Let $\mu^\star$ denote a second-best markup vector that finances fixed costs. Instead of measuring wedges relative to marginal-cost pricing, we can measure excess wedges relative to this benchmark:

$$\Delta \log \mu^{ex} = \log \mu - \log \mu^\star. \tag{25}$$

The aggregate welfare loss relative to the second-best benchmark is then computed by replacing $\Delta \log \mu$ with $\Delta \log \mu^{ex}$ in the quadratic welfare formula:

$$\mathscr{L}^{ex} \approx \frac{1}{2} \sum_j \lambda_j \theta_j \operatorname{Var}_{\Omega(j)} \left(\Psi \Delta \log \mu^{ex} \right). \tag{26}$$

Similarly, firm-level marginal effects and finite single-firm welfare gains can be computed using the same

curvature matrix, but evaluated with excess wedges. The excess marginal welfare effect is

$$M_k^{ex} = \left(G\Delta \log \mu^{ex}\right)_k, \tag{27}$$

and the corresponding finite single-firm welfare gain from setting firm k's excess wedge to zero is

$$\Delta TWC_k^{ex} = \Delta \log \mu_k^{ex} \cdot M_k^{ex} - \frac{1}{2}\left(\Delta \log \mu_k^{ex}\right)^2 G_{kk}. \tag{28}$$

These expressions show that fixed costs change the reference wedge vector, not the basic structure of welfare accounting. The empirical analysis therefore compares the baseline marginal-cost benchmark with several Ramsey benchmarks that differ in how the required fixed-cost-financing reference markup is constructed. Appendix G provides the details of this construction.

3 Data

This section describes the administrative data used to construct the empirical counterparts of the theory objects: firm-level markup wedges, the firm-to-firm production network, final-demand exposure, Domar weights, and the samples used for aggregate and firm-level welfare calculations. The empirical analysis requires observing both firms' pricing wedges and the network through which those wedges propagate. The Belgian administrative data are well suited for this purpose because they combine firm balance-sheet information with transaction-level data on domestic firm-to-firm trade. Section 3.1 describes the data sources, Section 3.2 defines the samples, and Sections 3.3 and 3.4 map the raw data into the objects entering the welfare formulas.

3.1 Data Sources

Our empirical analysis combines three administrative datasets collected by the National Bank of Belgium. The first is the annual accounts database, which contains firm balance-sheet and income-statement information. The second is the international trade database, which records firm-level imports and exports. The third is the universe of domestic business-to-business transactions recorded through the Belgian VAT system.

The core of the analysis is the VAT-based B2B transactions database described in Dhyne, Petrin, et al. (2015). This dataset records domestic transactions between VAT-registered firms when the annual transaction value between a seller and a buyer is at least EUR 250. Each observation identifies a seller, a buyer, a year, and

the value of transactions between the two firms. The data therefore allow us to observe the reported domestic production network directly at the firm level rather than infer it from sectoral input–output tables. Throughout the analysis, unreported transactions below the VAT threshold and transactions with non-domestic counterparties enter through firms' residual sales and input-use measures rather than as separately identified domestic links.

The Belgian firm-to-firm transaction data are a well-established empirical setting for studying production networks, firm heterogeneity, and imperfect competition. Magerman et al. (2016) use these data to study the micro origins of aggregate fluctuations. Bernard et al. (2022) use Belgian production-network data to study the origins of firm heterogeneity. Dhyne, Liubich, and Magerman (2017) study price dispersion within firm pairs, and Dhyne, Kikkawa, and Magerman (2022) use the same broad data environment to study imperfect competition in firm-to-firm trade. More broadly, firm-level production-network data have become central to the empirical study of production linkages (Atalay et al., 2011). Our analysis builds on this accepted data environment but constructs distinct welfare objects tailored to the general-equilibrium framework in Section 2.

We merge the B2B transactions with annual accounts and international trade data. The merged dataset contains information on sales, intermediate input purchases, employment, wage payments, tangible assets, value added, exports, and imports. These variables allow us to construct the revenue-based network matrix Ω, final-demand shares, Domar weights, markup wedges, and the curvature objects entering the welfare calculations.

3.2 Sample Definition

The sample covers the period 2002–2014, during which coverage of the transaction data is stable and consistent. We restrict attention to private, non-financial firms with positive operating activity. In particular, firms must report positive sales, positive employment, and positive tangible assets. These restrictions remove dormant entities and ensure that cost shares, production-network exposures, and markup measures are economically meaningful.

We also trim the top 1 percent and bottom 1 percent of markup observations within each year. This is standard in production-based markup measurement and limits the influence of accounting irregularities or extreme cost-share realizations. Because the welfare formulas are quadratic in markup wedges, extreme markup observations can mechanically generate large welfare contributions. Trimming therefore ensures that the results are not driven by outliers.

The empirical analysis uses two related samples. Aggregate TWC is computed using the full sample of firms satisfying the baseline data requirements. This allows the aggregate welfare measure to capture the broadest possible set of production linkages and final-demand exposures in the Belgian economy. Firm-level marginal

and finite welfare objects are computed for the largest 10,000 firms by sales. This restriction is imposed for computational reasons: the firm-level objects require manipulating high-dimensional curvature terms, whereas aggregate TWC can be computed on the broader network. Section 4.2.2 shows that this group accounts for 85.4 percent of full-sample welfare losses, 86.3 percent of sales, and 80.2 percent of value added in the baseline year, 2002. The firm-level analysis therefore focuses on the economically most relevant part of the firm distribution while preserving most of the aggregate welfare accounting.

Table 1 summarizes the main sample restrictions and measurement choices used throughout the empirical analysis.

Table 1: Sample and Measurement Choices

Item	Definition
Sample period	2002–2014.
Transaction data	Domestic VAT firm-to-firm transactions between VAT-registered firms. A seller-buyer link is observed when annual transactions between the two firms are at least EUR 250.
Baseline firm restrictions	Private, non-financial firms with positive sales, employment, and tangible assets.
Full aggregate sample	All firms satisfying the baseline data requirements; the complete domestic network contains about 100,000 firms. Aggregate TWC is computed on this sample.
Firm-level welfare sample	Largest 10,000 firms by sales. This sample is used to compute marginal welfare effects and finite single-firm welfare gains. In the baseline year, 2002, it accounts for 85.4 percent of full-sample TWC, 86.3 percent of sales, and 80.2 percent of value added.
Markup measure	Production-based markups using COGS as the variable input: $\mu_{it} = \beta^{COGS}_{s(i)} / \alpha^{COGS}_{it}$.
Markup trimming	Top 1 percent and bottom 1 percent of markup observations trimmed within each year.
Network objects	Revenue-based firm-to-firm input shares, final-demand exposure, Domar weights, and Leontief inverse constructed separately by year.

Notes: The VAT transaction threshold follows the data description in Dhyne, Petrin, et al. (2015). The top-10,000 coverage shares refer to the 2002 firm-only truncation exercise in Table 2.

3.3 Production Network Construction

We construct the revenue-based input–output matrix using observed nominal transaction flows. For each buyer i and seller j, we measure purchases by buyer i from seller j and divide them by buyer i's total sales. This yields the firm-level input–output matrix Ω defined in Section 2.1. Rows index buyers and columns index sellers, so each row describes the domestic supplier composition of a buyer's production costs measured at observed prices.

Because the denominator is sales, the row sum is not constrained to equal one; the remaining share reflects value added, imported inputs, and other input or expenditure components outside the observed domestic firm-to-firm network.

The associated Leontief inverse, $\Psi = (I - \Omega)^{-1}$, captures direct and indirect production linkages. It summarizes how a markup wedge at one firm propagates through the network to downstream buyers. Because the transaction data identify supplier–buyer links at the firm level, these propagation terms are constructed directly from observed firm-to-firm relationships.

We also construct final-demand exposure. Final demand is measured as sales outside the domestic firm-to-firm network, including household demand, exports, and other sales not recorded as domestic B2B purchases. Thus the observed B2B links discipline intermediate-input propagation, while residual sales discipline exposure to final demand and foreign or unobserved buyers. These final-demand shares determine the revenue-based Domar weights through the accounting identity $\lambda' = b'\Psi$. The resulting Domar weights measure each firm's macroeconomic exposure in the production network and provide the scale terms entering the welfare formulas.

All network objects are constructed separately by year. The production network, final-demand exposure, Domar weights, and welfare objects are therefore allowed to vary over time.

3.4 Markup Measurement

Firm-level markups are estimated using the production approach of De Loecker and Warzynski (2012). Under cost minimization, the markup equals the ratio of a variable input's output elasticity to its revenue share. We use costs of goods sold (COGS) as the variable input and compute firm-level markups as

$$\mu_{it} = \frac{\beta_{s(i)}^{COGS}}{\alpha_{it}^{COGS}},$$

where α_{it}^{COGS} is firm i's COGS share in revenue in year t, and $\beta_{s(i)}^{COGS}$ is the output elasticity assigned to firm i's sector.

Following the approach in De Loecker, Eeckhout, and Unger (2020), we construct the sector-level output elasticity from the median COGS revenue share within each sector. This procedure allows the output elasticity to vary across sectors rather than imposing a common calibrated value for all firms. The resulting markup compares each firm's COGS share to a sector-specific benchmark cost share: firms with lower variable-input shares relative to their sector benchmark receive higher measured markups. The welfare analysis therefore uses

within-sector variation in variable-input shares, together with sector differences in the benchmark elasticity, to construct firm-level markup wedges.

The same markup measure is used throughout the aggregate and firm-level welfare calculations. Additional details on the construction of markups, network matrices, cost-based objects, and welfare measures are reported in Appendix B.7.

4 Empirical Results

This section turns the welfare formulas into empirical objects and asks three questions. How large is the aggregate welfare cost of markup distortions? How much of that cost is shaped by production-network propagation? And which firms account for the welfare-relevant distortions beneath the aggregate loss? We first compare aggregate TWC in the baseline production-network economy with a horizontal benchmark. We then compute firm-level marginal welfare effects, M_k, and finite single-firm welfare gains, ΔTWC_k. Finally, we test whether high-impact firms are persistent over time and whether their identity is robust to Ramsey benchmarks that allow markups to finance fixed costs.

The empirical results deliver a common message. Markup distortions are costly in the aggregate, but their welfare consequences are highly uneven across firms. A small upper tail accounts for a large share of welfare impact, and those high-impact firms remain important across years and benchmark economies. The empirical task is therefore not only to measure aggregate misallocation, but also to identify which firms account for welfare-relevant markup distortions.

4.1 Markups and Network Structure

The welfare calculations are disciplined by two empirical primitives: firm-level markup wedges and firm-to-firm production linkages. Markup dispersion determines the size of firm-level wedges, while the production network determines how these wedges propagate to downstream buyers and final demand.

The pooled distribution of firm-level markups is strongly right-skewed: most firms charge markups moderately above marginal cost, while a smaller set of firms exhibits substantially higher markups. Appendix A reports the markup distribution and average markups under alternative weighting schemes. This heterogeneity generates markup wedges that are necessary for welfare losses, but not sufficient to identify the firms that matter most. Welfare relevance depends on markup wedges interacted with economic exposure, network position, and the

reallocation induced by a markup change.

Figure 1 shows the evolution of average markups between 2002 and 2014. Across weighting schemes, markups increase over time, although the magnitude of the increase differs depending on the weights used. The welfare analysis below asks how changes in markup wedges translate into aggregate losses once network propagation and substitution across suppliers are taken into account.

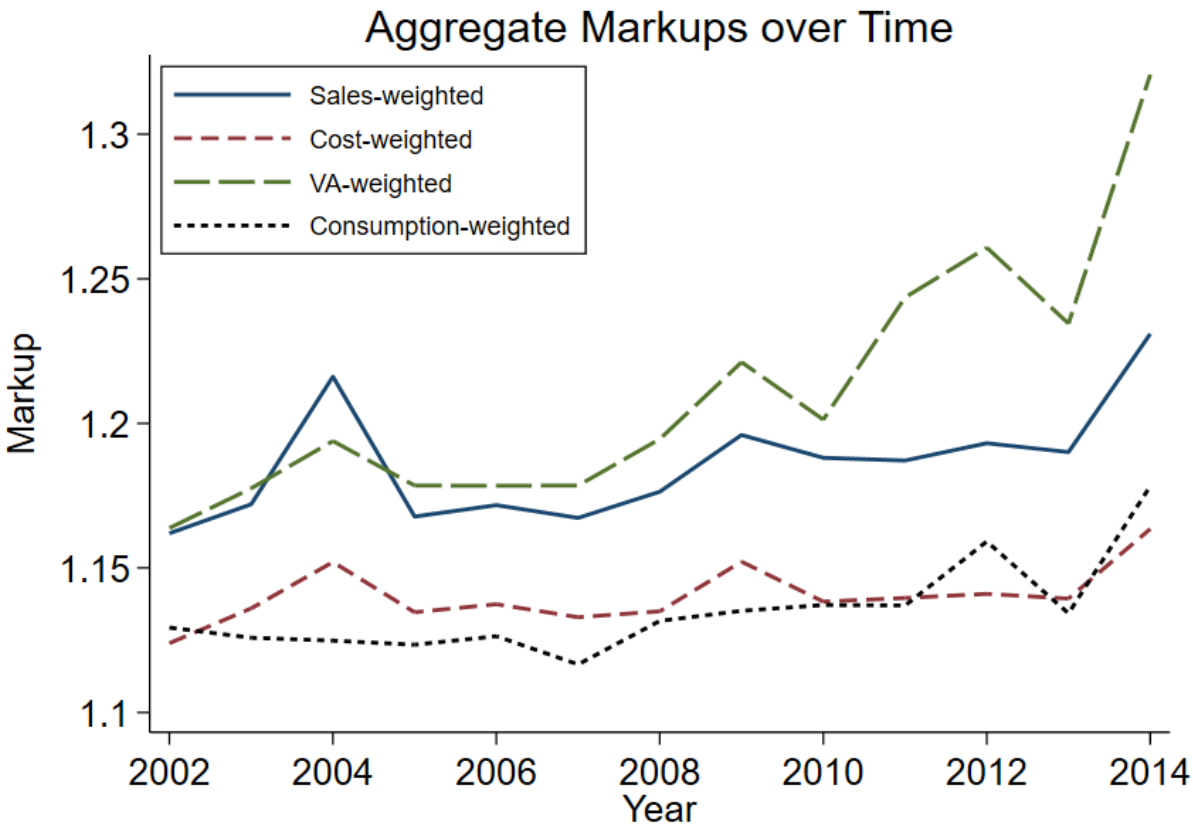


Figure 1: Average Markups by Weighting Scheme, 2002–2014

The second primitive is the firm-to-firm production network. The Belgian B2B data reveal a highly connected economy in which firms are linked through supplier–buyer relationships. These links imply that a markup affects not only the firm that charges it, but also downstream producers that use its output as an intermediate input. The next subsection combines markup wedges and network linkages to measure aggregate welfare losses.

4.2 Aggregate Welfare Costs and Network Amplification

Aggregate welfare losses rise substantially over the sample period, and production linkages amplify their level. We quantify these losses using the quadratic representation derived in Section 2. TWC summarizes the output

loss associated with observed markup wedges, taking into account both their dispersion and their propagation through the firm-to-firm production network.

4.2.1 Network Versus Horizontal Economy

To isolate the role of input–output linkages, we compare the baseline network economy to a horizontal benchmark in which firm-to-firm intermediate-input propagation is shut down. In the horizontal economy, firms do not use intermediate inputs from one another. Welfare losses therefore reflect dispersion in firm-level markup wedges across final demand rather than dispersion in propagated wedges across buyers' input bundles.

Figure 2 compares aggregate welfare losses in the baseline network economy and in the horizontal benchmark.

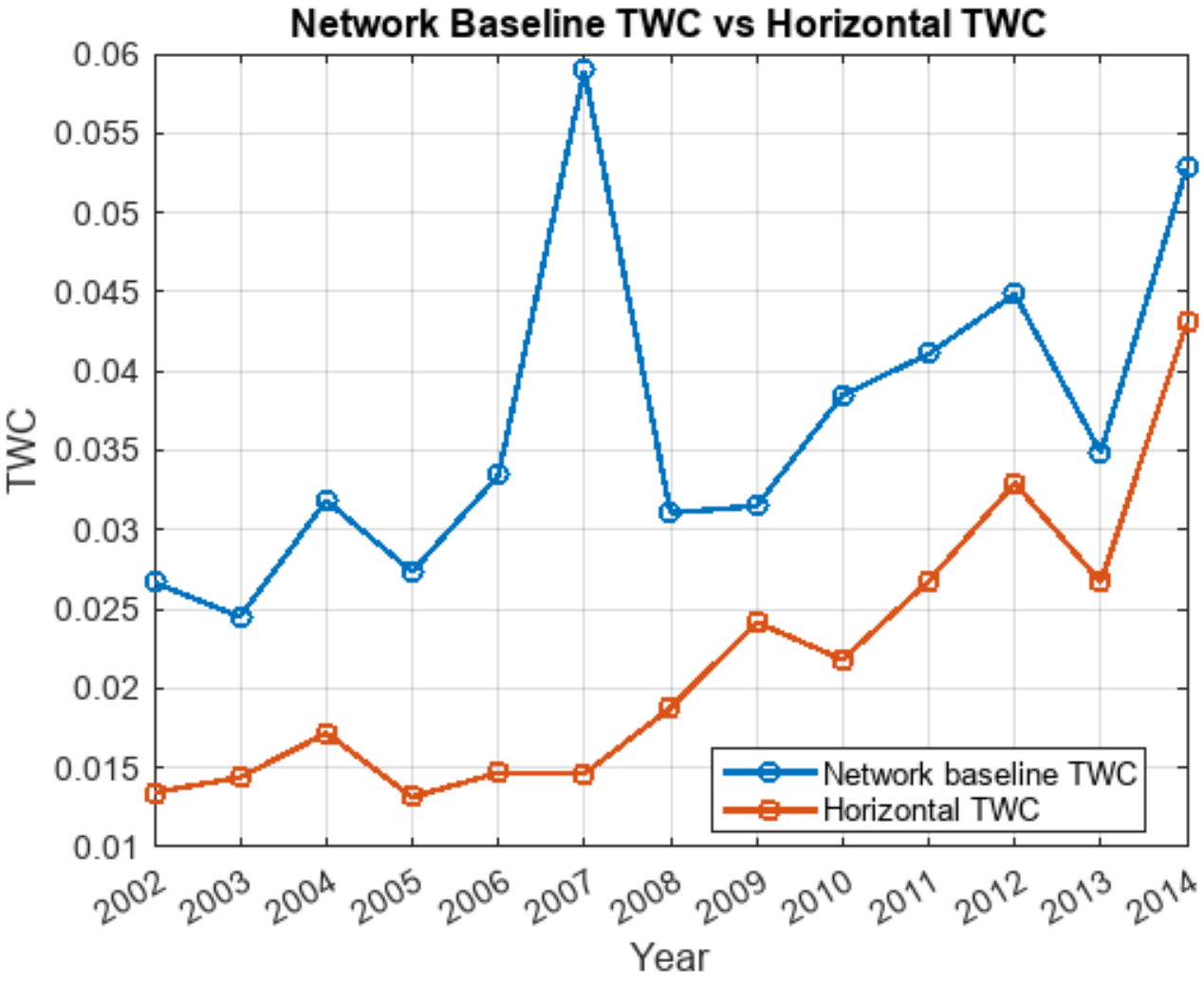


Figure 2: Aggregate TWC in the Baseline Network and Horizontal Benchmark

In the baseline network economy, TWC rises from 0.0267 log points in 2002 to 0.0528 log points in 2014.

The measured welfare cost of markups therefore approximately doubles over the sample period. Aggregate losses are larger in the baseline network economy than in the horizontal benchmark, showing that production linkages amplify the welfare cost of markup dispersion. At the same time, the horizontal benchmark follows a similar time-series pattern. The increase in welfare losses is therefore primarily associated with changes in markup wedges, while the production network raises the level of losses by transmitting those wedges through input–output linkages.

The remaining analysis focuses on the baseline network economy. Benchmark exercises are used to assess how much the level of measured losses, and the identity of high-impact firms, depend on the reference environment.

4.2.2 Truncated Economies

Aggregate losses are concentrated among large firms, but measuring this concentration requires preserving their interactions with the rest of the economy. To show this, we construct truncated versions of the economy for the baseline year 2002. The exercise separates two forces: the direct role of large firms and the role of their interactions with the long tail of smaller firms.

We consider two complementary truncation exercises.

Firm-only truncation. The first exercise retains only the top N firms ranked by sales and removes all other firms from the network. We then recompute TWC on this reduced economy. This exercise asks how much aggregate misallocation is captured by the largest firms when their interactions with omitted firms are ignored.

Table 2 reports the results.

Table 2: Firm-Only Truncation of Aggregate TWC

N	$L(N)$	% TWC	Sales (%)	VA (%)	Labor (%)
10	0.0004	1.5	16.7	16.9	14.1
100	0.0096	36.0	36.4	34.4	30.4
1 000	0.0154	57.7	61.9	57.7	55.9
10 000	0.0228	85.4	86.3	80.2	81.0
15 000	0.0236	88.4	89.6	84.0	85.3
20 000	0.0240	89.9	92.3	86.8	88.3
Full sample	0.0267	100	100	100	100

Notes: The table recomputes aggregate TWC in 2002 after retaining only the top N firms by sales and removing all other firms from the network. Columns report the resulting welfare loss, denoted by $L(N)$, and the retained firms' shares of full-sample TWC, sales, value added, and labor.

The firm-only truncation reveals substantial concentration. The top 100 firms account for 36.0 percent of full-sample TWC, while the top 1,000 firms account for 57.7 percent. The top 10,000 firms account for 85.4 percent of aggregate welfare losses. These shares are broadly comparable to, but not identical to, the corresponding sales and value-added shares reported in Table 2. Large firms account for a disproportionate share of aggregate misallocation, but welfare relevance is not mechanically equivalent to economic size.

At the same time, excluding smaller firms entirely understates aggregate welfare losses. Even when the top 20,000 firms are retained, the reduced economy captures 89.9 percent of full-sample TWC. The remaining firms are individually small, but collectively they account for a non-negligible part of the welfare loss.

Residual-node truncation. The second exercise retains the top N firms but aggregates all omitted firms into a single residual node, interpreted as the rest of the economy. This node preserves the aggregate input and output relationships between the top firms and the omitted firms, while abstracting from heterogeneity within the residual group. The residual node's markup is constructed as a total-cost-weighted average of the markups of the omitted firms.

Appendix Table 7 reports the results. Including the residual node increases measured welfare losses relative to the firm-only truncation, especially when N is small. With only the top 10 firms, the firm-only economy captures 1.5 percent of full-sample TWC, whereas the residual-node economy captures 33.3 percent. With the top 100 firms, the corresponding shares are 36.0 percent and 52.1 percent. The gap narrows as more firms are retained explicitly, but it remains informative about the role of the long tail.

This comparison shows that aggregate misallocation is shaped by both concentration and network interaction. Large firms account for a substantial share of welfare losses, but their welfare impact cannot be measured accurately by isolating them from the rest of the economy. The long tail matters because it supplies inputs to, buys inputs from, and otherwise interacts with the largest firms. Aggregating the long tail into a residual node preserves these aggregate linkages and recovers part of the amplification lost in the firm-only truncation, as shown by the gap between Appendix Table 7 and Table 2.

Summary. The aggregate results deliver three conclusions. First, the welfare cost of markups increases substantially over the sample period. Second, production linkages amplify welfare losses relative to the horizontal benchmark. Third, aggregate misallocation is concentrated among large firms, but the relevant object combines size, markup wedges, and network interaction. These facts motivate the firm-level analysis below: aggregate TWC measures how large misallocation is, but not which firms account for the greatest welfare leverage.

4.3 Firm-Level Welfare Impact

Aggregate TWC measures the level of misallocation in the economy, but it does not identify which firms matter most for welfare. We therefore turn from the size of the aggregate loss to its firm-level attribution. The two objects derived in Section 2, the marginal welfare effect M_k and the finite single-firm welfare gain ΔTWC_k, answer related but distinct ranking questions.

The distinction between aggregate losses and firm-level welfare impact is central to the paper. A firm can be large or charge a high markup without having the largest welfare effect from either a marginal or a finite markup-wedge reduction. Both M_k and ΔTWC_k depend on markup wedges, economic exposure, production-network position, and the reallocation induced by a markup change.

Throughout the firm-level analysis, we focus on the largest 10,000 firms by sales. As shown in Section 4.2.2, this group captures a large majority of full-sample welfare losses while keeping the computation of firm-level objects tractable. The resulting sample covers the economically relevant upper part of the firm distribution while retaining most of the aggregate welfare accounting.

4.3.1 Marginal Welfare Effects

The marginal welfare effect M_k measures local welfare leverage: the change in aggregate welfare losses induced by a marginal change in firm k's markup wedge, holding other markup wedges fixed.

The distribution of baseline marginal welfare effects is highly right-skewed. Most firms have very small marginal effects, while a small upper tail exhibits substantially larger welfare leverage. Appendix A reports the underlying distribution of M_k. This concentration has a direct reform-ranking interpretation: rankings based on M_k identify firms with the largest local welfare leverage, incorporating network propagation and general-equilibrium reallocation rather than relying on sales or markups alone.

We next examine finite single-firm reforms to assess whether firms with large local marginal effects also generate large welfare gains when their markup wedges are reduced discretely.

4.3.2 Finite Single-Firm Reforms

Many policy interventions are discrete rather than infinitesimal. We therefore complement the marginal analysis with finite single-firm reforms.

For each firm k, we compute the welfare gain from setting its markup wedge to the benchmark while holding all other firms' markup wedges fixed. We denote this object by ΔTWC_k. As derived in Section 2.4, the finite

single-firm gain is

$$\Delta TWC_k = \Delta \log \mu_k \cdot M_k - \frac{1}{2}(\Delta \log \mu_k)^2 G_{kk}. \tag{29}$$

where M_k is the marginal welfare effect and G_{kk} is the curvature of the welfare loss with respect to firm k's markup wedge. The first term captures the gain implied by the initial marginal effect. The second term adjusts for the fact that the marginal effect changes as the wedge is reduced.

The distribution of baseline finite single-firm welfare gains is also highly skewed. Most firms generate very small finite single-firm gains when considered in isolation, while a small set of firms generates much larger gains. Appendix A reports the underlying distribution of ΔTWC_k. The concentration of welfare-relevant distortions is therefore not merely a local property of the marginal derivative; it also appears when evaluating discrete single-firm reforms.

4.3.3 Relationship Between Marginal and Finite Measures

Figure 3 compares the concentration of positive finite single-firm welfare gains and positive marginal welfare effects.

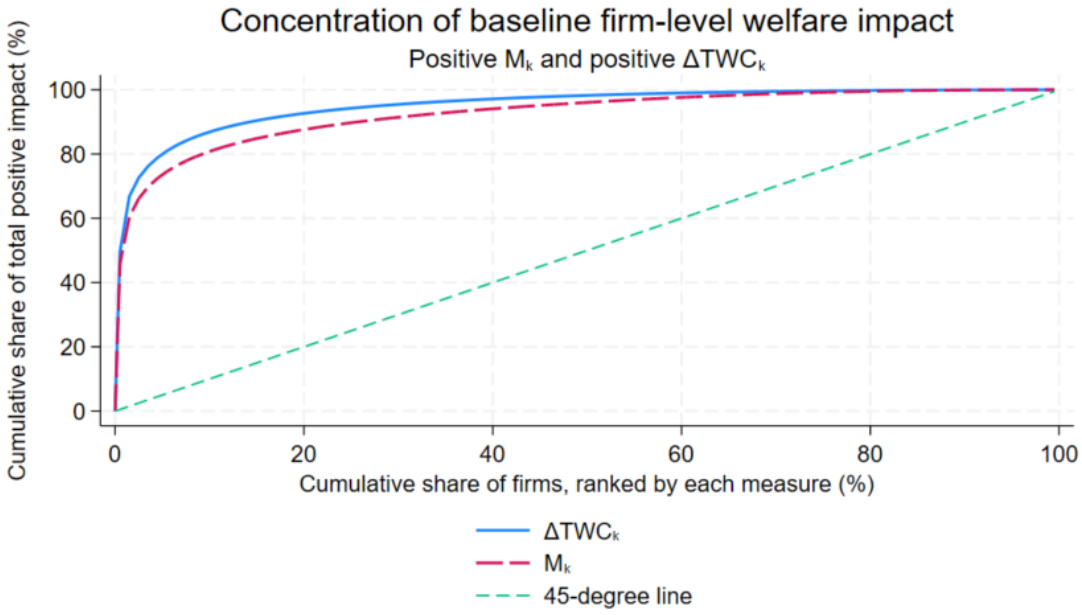


Figure 3: Concentration of Firm-Level Welfare Impact

The figure shows that both measures are highly concentrated, indicating that firm-level welfare impact is concentrated whether one evaluates local marginal changes or discrete single-firm reforms. The concentration is not just a feature of the aggregate truncation exercise; it also appears in the firm-level welfare statistics that

rank potential markup-wedge reforms. At the same time, the two objects are conceptually distinct. The marginal effect M_k measures the slope of welfare losses at the observed allocation, whereas ΔTWC_k integrates this slope over a discrete reform path and subtracts the curvature correction.

Table 3 reports the Spearman rank correlation between the marginal and finite single-firm welfare measures.

Table 3: Rank Correlation Between Marginal and Finite Firm-Level Welfare Measures

Variables	Spearman's ρ
M_k^{Network} and $\Delta TWC_k^{\text{Baseline}}$	0.5844

Notes: The table reports the Spearman rank correlation between the marginal welfare effect, M_k^{Network}, and the finite single-firm welfare gain, $\Delta TWC_k^{\text{Baseline}}$, computed at the firm level. The null hypothesis of independence is rejected at the 1 percent level.

The rank correlation is positive and statistically significant, showing that marginal welfare effects contain useful information about finite single-firm reform gains. At the same time, the correlation is not perfect. Local welfare leverage and discrete single-firm welfare gains are closely related, but they are not interchangeable. We therefore report persistence and robustness results for both M_k and ΔTWC_k.

4.4 Persistence of Firm-Level Welfare Impact

Firm-level welfare impact is highly concentrated. We now ask whether the identity of high-impact firms is persistent over time. This distinction matters for interpretation. If high-impact firms changed from year to year, the concentration results would mainly reflect transitory shocks. If the same firms repeatedly appear at the top of the distribution, firm-level welfare impact instead reflects persistent features of firms, such as their size, markups, customer base, and position in the production network.

We examine persistence using top-group survival rates in the main text. Appendix A reports complementary rank-persistence regressions and transition matrices across quintiles.

Table 4 focuses directly on firms in the upper tail of the distribution. It reports the probability that firms in the top 10 percent, top 5 percent, and top 1 percent remain in the same group one year later.

The upper tail is highly stable. For ΔTWC_k, about 70–72 percent of observed top-group firms remain in the same top group one year later. For M_k, the corresponding conditional persistence rates are about 86–87 percent. The concentration documented above is therefore not driven by a rotating set of temporary outliers.

Appendix Tables 8, 9, and 10 confirm the same pattern. Firms in the highest-impact part of the distribution are much more likely to remain high-impact than to move to the middle or bottom of the distribution.

Table 4: One-Year Persistence of High-Impact Firms

Measure	Group	Stay same group	Stay cond. observed	Observed next year	Exit sample	Fall out cond. observed
ΔTWC_k	Top 10%	64.7	72.4	89.4	10.6	24.7
ΔTWC_k	Top 5%	63.2	70.5	89.7	10.3	26.5
ΔTWC_k	Top 1%	61.9	69.8	88.7	11.3	26.8
M_k	Top 10%	81.2	86.9	93.4	6.6	12.2
M_k	Top 5%	79.9	85.7	93.2	6.8	13.3
M_k	Top 1%	80.4	87.3	92.2	7.8	11.7

Notes: Entries are percentages. The table reports one-year persistence rates for firms in the top part of the within-year distribution of firm-level welfare impact. "Stay same group" counts firms that disappear from the sample in the following year as non-persistent. "Stay cond. observed" conditions on the firm being observed in the following year. "Fall out cond. observed" is the share of firms that are observed in the following year but no longer belong to the same top group. Origin years are 2002–2013.

Taken together, these results show that firm-level welfare impact is both concentrated and persistent. The firms that matter most for welfare are not merely temporary outliers; they occupy persistently important positions in the economy.

4.5 Fixed Costs and Ramsey Benchmarks

The baseline analysis evaluates observed markup wedges relative to marginal-cost pricing. This benchmark measures the allocative cost of observed markup wedges relative to the efficient allocation. If firms face fixed operating costs, however, strictly marginal-cost pricing may not be feasible. Some positive markups may be required to cover fixed costs, and the relevant welfare benchmark may be second-best rather than first-best.

Following Section 2.6, we construct Ramsey benchmarks that interpret part of observed markups as necessary to finance fixed costs. We then define excess markup wedges relative to these benchmarks and recompute welfare losses and firm-level welfare effects. This exercise asks whether the main empirical conclusions depend on treating all observed markup wedges as distortions relative to marginal-cost pricing.

We consider three Ramsey benchmarks. The first imposes a common economy-wide reference markup. The second allows the reference markup to vary across industries. The third allows the reference markup to vary at the firm level. These alternatives differ in how much heterogeneity they allow in the markup needed to finance fixed costs. The baseline specification remains the marginal-cost benchmark.

4.5.1 Aggregate Welfare Losses

Figure 4 compares aggregate welfare losses under the baseline network economy and the Ramsey benchmarks.

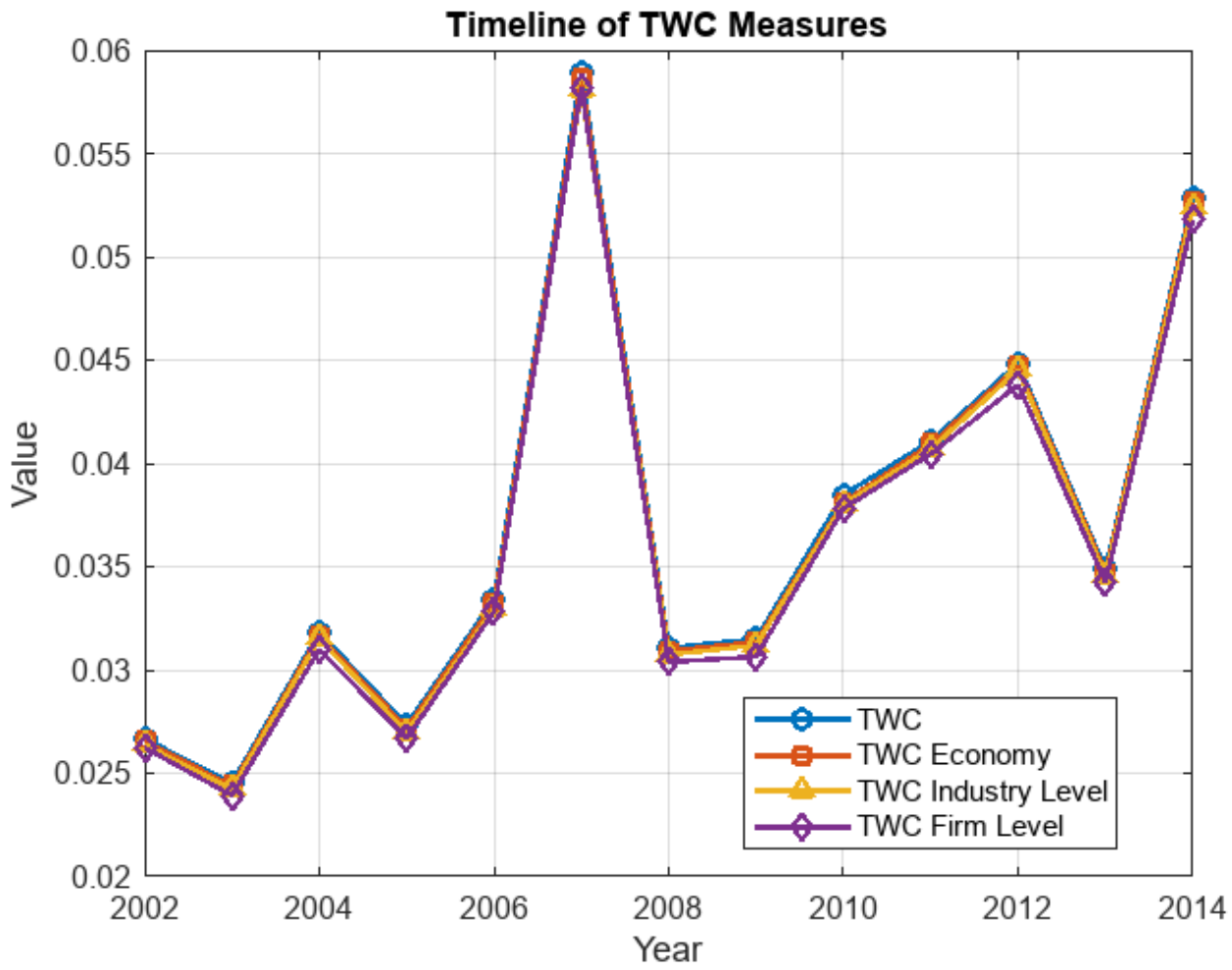


Figure 4: Aggregate TWC Under Baseline and Ramsey Benchmarks

The Ramsey benchmarks reduce measured welfare losses relative to the baseline network economy because part of the observed markup is interpreted as financing fixed costs rather than generating excess distortions. Nevertheless, excess welfare losses remain economically meaningful, and their evolution over time is closely related to the baseline measure in Figure 4. Fixed-cost considerations affect the level of measured misallocation, but they do not overturn the conclusion that markup distortions generate sizable aggregate welfare losses.

4.5.2 Firm-Level Rankings

We next examine whether the identity of high-impact firms changes when welfare is evaluated relative to Ramsey benchmarks. This is the key robustness check for the firm-level analysis. If the same firms remain important across benchmarks, then the concentration and persistence results are not an artifact of using marginal-cost pricing as the reference allocation.

Appendix Table 11 reports Spearman rank correlations across finite single-firm welfare gains in the firm-level ranking sample, comparing the baseline network economy with the Ramsey benchmarks. The rank correlations are very high across all benchmark definitions. Firms that generate large finite single-firm welfare gains in the baseline network economy also tend to generate large excess finite single-firm welfare gains under the economy-level, industry-level, and firm-level Ramsey benchmarks. Allowing for fixed costs changes the level of measured welfare losses much more than it changes the ranking of firms.

4.5.3 Rank Stability Among Top Firms

Rank correlations over the firm-level ranking sample are informative, but they may partly reflect the fact that many low-impact firms remain low-impact across specifications. For policy interpretation, the relevant question is whether the identity of firms in the upper tail is also stable. We therefore examine rank stability among the top firms in the welfare-impact distribution.

This exercise focuses on the firms that matter most for aggregate welfare. If the same firms appear in the upper tail across the baseline network economy and the Ramsey benchmarks, then the welfare ranking is robust to alternative assumptions about fixed-cost financing. Conversely, if top-firm rankings changed substantially across benchmarks, then the identity of high-impact firms would depend sensitively on the choice of reference allocation.

The top-firm results reinforce the ranking-sample evidence. Among firms in the top 1, 5, or 10 percent of the baseline ΔTWC_k distribution, rank correlations with the Ramsey benchmarks are close to one. Fixed-cost adjustments therefore do not merely preserve broad rank correlations; they also leave the welfare-relevant upper tail largely intact.

Taken together, the Ramsey results support a clear conclusion: fixed costs matter for the level of welfare accounting, but they do not substantially alter the identity of firms with the largest welfare impact. The concentration and persistence of firm-level welfare relevance remain central features of the data under both first-best and second-best benchmark economies.

Table 5: Rank Correlations for Top-Impact Firms Across Ramsey Benchmarks

	Top 1%	Top 5%	Top 10%
Panel A: Top firms ranked by baseline ΔTWC_k			
M_k vs. baseline ΔTWC_k	0.568	0.598	0.570
Baseline ΔTWC_k vs. Ramsey economy	0.999	0.997	0.997
Baseline ΔTWC_k vs. Ramsey industry	0.995	0.992	0.991
Baseline ΔTWC_k vs. Ramsey firm	0.999	0.984	0.975
Panel B: Top firms ranked by baseline M_k			
M_k vs. baseline ΔTWC_k	0.237	0.304	0.329
Baseline ΔTWC_k vs. Ramsey economy	0.997	0.997	0.998
Baseline ΔTWC_k vs. Ramsey industry	0.993	0.992	0.992
Baseline ΔTWC_k vs. Ramsey firm	1.000	0.999	0.998

Notes: Entries are Spearman rank correlations. Top groups are defined within year. Panel A restricts the sample to firms in the top 1, 5, or 10 percent of the baseline network-economy finite welfare-gain distribution, ΔTWC_k. Panel B restricts the sample to firms in the top 1, 5, or 10 percent of the baseline network-economy marginal welfare-effect distribution, M_k. Ramsey economy, Ramsey industry, and Ramsey firm refer to finite welfare-gain measures computed relative to the corresponding second-best benchmark.

5 Conclusion

This paper asks which firms matter most for the welfare cost of market power in a production network. Existing aggregate welfare measures quantify the total cost of distortions, but do not identify where welfare-relevant market power is located. This distinction is important because many policy interventions are local: they affect particular firms, markets, or supplier–buyer relationships rather than eliminating markup distortions economy-wide.

The central contribution of the paper is to move from aggregate welfare accounting to firm-level welfare attribution. Building on the quadratic welfare representation of Baqaee and Farhi (2020), we compute the Total Welfare Cost of markups and derive two firm-level objects. The marginal welfare effect, M_k, measures the local welfare impact of changing firm k's markup wedge. The finite single-firm welfare gain, ΔTWC_k, measures the gain from setting one firm's markup wedge to the relevant benchmark while holding all other wedges fixed. Both objects combine markup wedges, economic exposure, production-network position, and general-equilibrium reallocation.

The empirical analysis uses administrative firm-to-firm transaction data for Belgium from 2002 to 2014. It delivers three main findings. First, the aggregate welfare cost of markups rises substantially over the sample period, with baseline TWC roughly doubling from 0.0267 log points in 2002 to 0.0528 log points in 2014. Production linkages amplify these losses relative to a horizontal benchmark without firm-to-firm input propagation.

Second, truncation exercises show that large firms account for a substantial share of aggregate losses, but also that preserving interactions with the broader production network matters for measuring welfare impact.

Third, the firm-level results reveal substantial heterogeneity beneath the aggregate welfare loss. Marginal welfare effects and finite single-firm welfare gains are both highly concentrated: a small upper tail of firms accounts for a large share of welfare-relevant distortions. These high-impact firms are persistent over time, indicating that the upper tail is not a rotating set of temporary outliers. The identity of high-impact firms is also robust to Ramsey benchmarks that allow part of observed markups to finance fixed costs. Fixed-cost adjustments affect the level of measured welfare losses more than they affect firm-level rankings.

These findings imply that sales and markups, taken alone, are incomplete guides to welfare relevance. A firm's welfare impact depends on how its markup wedge interacts with economic exposure, production-network position, and the reallocation induced by a markup change. Production networks therefore shape not only the aggregate welfare cost of market power, but also the identity of the firms that matter most for welfare. The main implication is that welfare-based analysis of market power should rank firms by their contribution to network-wide misallocation, not only by the size of their markup or their footprint in sales.

The analysis is conditional on observed firms, technologies, markups, and production linkages. Future work could study how firm-level interventions affect entry, exit, innovation, and endogenous supplier–buyer relationships. Another direction is to compare high-impact firms across countries or institutional environments. The main message is that understanding the welfare cost of market power requires knowing not only how large markup distortions are, but also where those distortions occur in the production network.

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Appendix Roadmap

The appendix has three roles. Appendix A reports supporting empirical results for the markup distribution, truncation exercises, firm-level concentration, persistence, and Ramsey robustness.

Appendix B documents the measurement and calibration choices used to construct firm-level markups, production-network objects, Domar weights, and the welfare objects used in the empirical analysis.

Appendices C–G provide the theoretical details behind the main text. Appendix C derives the quadratic representation of aggregate welfare losses. Appendix D derives marginal welfare effects and finite single-firm reform formulas. Appendix E clarifies the local interpretation and possible second-best sign ambiguity of marginal effects. Appendix F develops the horizontal economy benchmark, and Appendix G extends the framework to fixed costs and Ramsey reference markups.

A Additional Empirical Results

This appendix collects supporting empirical results that complement the main text. The figures and tables below document the primitives used in the welfare calculations and provide additional evidence for the aggregate, concentration, persistence, and robustness results reported in Section 4.

A.1 Markup Distribution and Average Markups

The main text summarizes the evolution of average markups over time. This subsection reports two complementary facts: the pooled distribution of firm-level markups and average markups under alternative weighting schemes. These results document the heterogeneity in markup wedges that enters the welfare calculations.

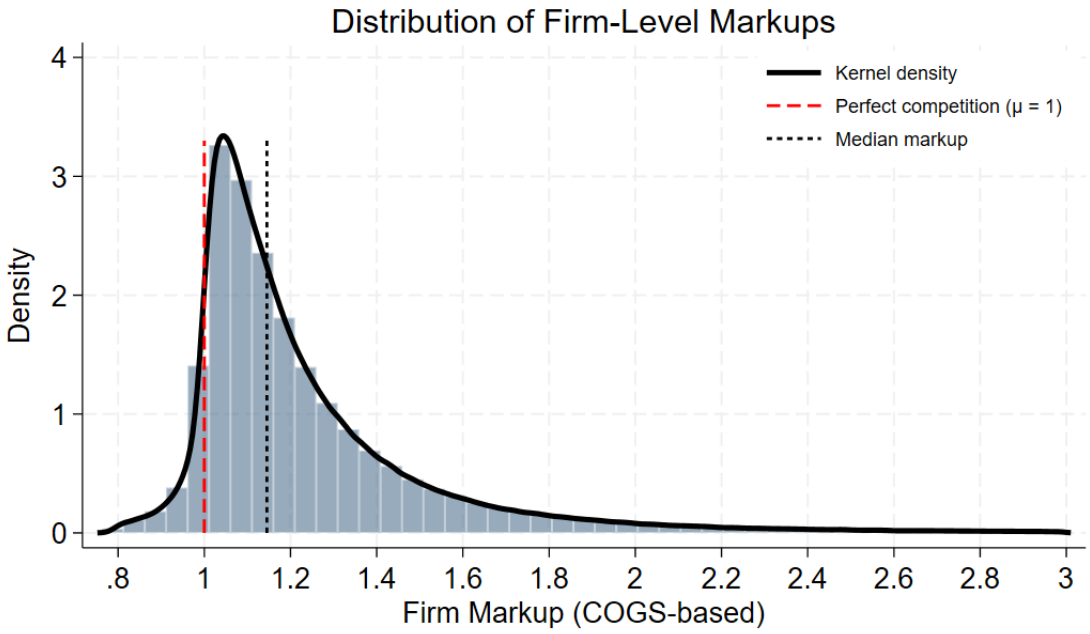


Figure 5: Distribution of Firm-Level Markups, 2002–2014

The figure pools firm-year observations after the annual 1/99 percent markup trimming rule used in the welfare calculations.

Table 6 reports average markups under alternative weighting schemes.

Table 6: Average Markups by Weighting Scheme

	Arithmetic	Sales	Final sales	Cost	Value added
Average markup	1.25	1.19	1.14	1.14	1.21

Notes: The table reports averages over the 2002–2014 firm-year sample after annual 1/99 percent markup trimming. Final sales weights use sales outside the observed domestic B2B network, including household demand, exports, and other residual sales.

A.2 Residual-Node Truncation

The main text reports the firm-only truncation exercise and discusses the residual-node version. Table 7 reports the full residual-node results. The exercise retains the top N firms explicitly and aggregates omitted firms into a single residual node that preserves aggregate links between the retained firms and the rest of the economy.

Table 7: Residual-Node Truncation of Aggregate TWC

N	$L^R(N)$	% of Full TWC	Δ vs Firm-only
10	0.0089	33.3	+0.0085
100	0.0139	52.1	+0.0043
1 000	0.0196	73.4	+0.0042
10 000	0.0232	86.9	+0.0004
15 000	0.0238	89.1	+0.0002
20 000	0.0242	90.6	+0.0002
Full sample	0.0267	100	

Notes: The table recomputes aggregate TWC in 2002 after retaining the top N firms by sales and aggregating omitted firms into a residual node. $L^R(N)$ denotes the residual-node truncation estimate. The last column reports the difference relative to the firm-only truncation in Table 2.

A.3 Firm-Level Distributions and Concentration

The main text reports a combined concentration figure for marginal welfare effects and finite single-firm welfare gains. This subsection reports the underlying distributions, documenting the upper tail behind the concentration result.

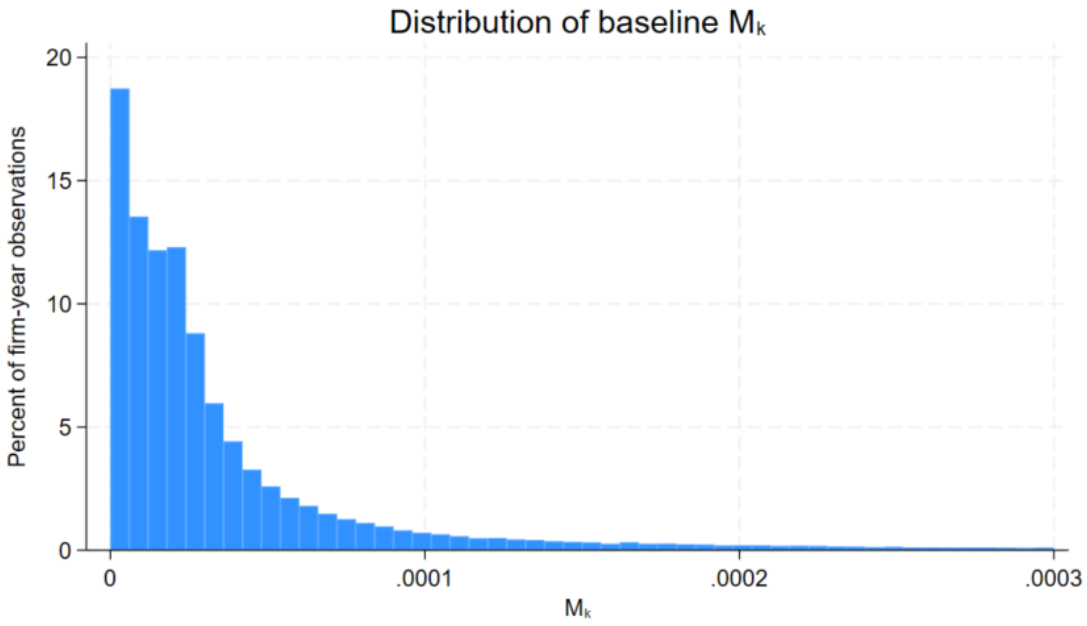


Figure 6: Distribution of Baseline Marginal Welfare Effects

The figure plots M_k for the firm-level ranking sample in the baseline network economy; larger values indicate larger local welfare leverage.

Figure 7 plots the distribution of baseline finite single-firm welfare gains.

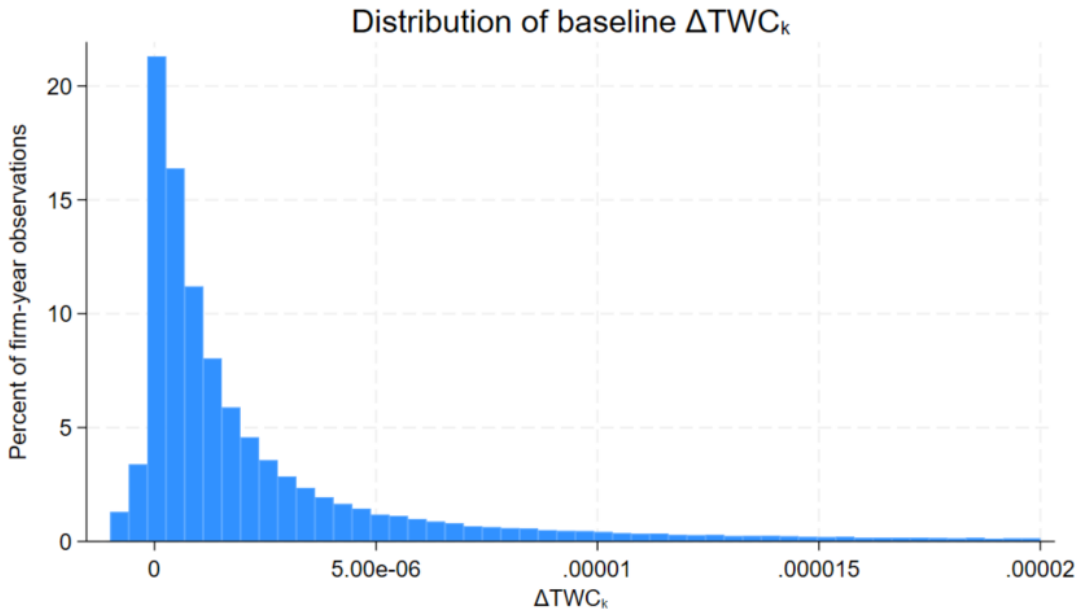


Figure 7: Distribution of Baseline Finite Single-Firm Welfare Gains

The figure plots ΔTWC_k for the firm-level ranking sample in the baseline network economy; larger values indicate larger finite single-firm welfare gains.

A.4 Persistence Checks

The main text reports top-group survival rates to summarize persistence in the upper tail of the welfare-impact distribution. This subsection reports two additional checks: rank-persistence regressions and one-year transition matrices across quintiles.

Tables 9 and 10 report one-year transition matrices across quintiles for ΔTWC_k and M_k, respectively.

A.5 Ramsey Benchmark Rank Correlations

The main text reports rank stability among top-impact firms. Table 11 reports the corresponding Spearman rank correlations for finite single-firm welfare gains in the firm-level ranking sample under the baseline network economy and the Ramsey benchmarks.

Table 8: Rank Persistence of Firm-Level Welfare Impact

	ΔTWC_k		M_k	
Dependent variable: percentile rank in year t	(1)	(2)	(3)	(4)
Lag 1 rank	0.741***	0.484***	0.918***	0.658***
	(0.003)	(0.007)	(0.002)	(0.008)
Lag 2 rank		0.181***		0.157***
		(0.007)		(0.009)
Lag 3 rank		0.113***		0.064***
		(0.007)		(0.008)
Lag 4 rank		0.092***		0.080***
		(0.006)		(0.006)
Cumulative persistence	0.741	0.870	0.918	0.960
Year fixed effects	Yes	Yes	Yes	Yes
Clustered SE	Firm	Firm	Firm	Firm
R^2	0.541	0.608	0.835	0.868

Notes: The dependent variable is the within-year percentile rank of the indicated baseline welfare-impact measure for the firm-level ranking sample, with higher ranks denoting larger welfare impact. Columns (1) and (3) include one lag; columns (2) and (4) include four lags. All specifications include year fixed effects. Standard errors, clustered by firm, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 9: One-Year Transition Matrix for Finite Single-Firm Welfare Gains

	Quintile in year $t+1$				
Quintile in year t	Q1	Q2	Q3	Q4	Q5
Q1	64.8	18.1	7.5	5.0	4.6
Q2	19.4	52.1	20.3	5.9	2.2
Q3	7.4	20.7	47.8	20.1	4.0
Q4	4.8	6.0	19.7	54.2	15.3
Q5	4.7	2.0	3.7	14.5	75.1

Notes: Entries are transition probabilities in percent for baseline finite single-firm welfare gains, ΔTWC_k, in the firm-level ranking sample. Quintiles are computed within year; Q1 denotes the lowest-impact quintile and Q5 the highest-impact quintile. Origin years are 2002–2013.

Table 10: One-Year Transition Matrix for Marginal Welfare Effects

	Quintile in year $t+1$				
Quintile in year t	Q1	Q2	Q3	Q4	Q5
Q1	82.7	12.9	2.2	1.4	0.8
Q2	13.0	66.3	16.3	3.2	1.1
Q3	2.4	14.7	63.4	18.0	1.5
Q4	1.5	3.2	14.0	70.6	10.8
Q5	0.7	0.9	1.3	9.5	87.6

Notes: Entries are transition probabilities in percent for baseline marginal welfare effects, M_k, in the firm-level ranking sample. Quintiles are computed within year; Q1 denotes the lowest-impact quintile and Q5 the highest-impact quintile. Origin years are 2002–2013.

Table 11: Rank Correlations Across Firm-Level Welfare Effects

	$\Delta TWC_k^{\text{Baseline}}$	$\Delta TWC_k^{\text{Economy}}$	$\Delta TWC_k^{\text{Industry}}$	$\Delta TWC_k^{\text{Firm}}$
$\Delta TWC_k^{\text{Baseline}}$	1.0000			
$\Delta TWC_k^{\text{Economy}}$	0.9973	1.0000		
$\Delta TWC_k^{\text{Industry}}$	0.9884	0.9897	1.0000	
$\Delta TWC_k^{\text{Firm}}$	0.9911	0.9882	0.9796	1.0000

Notes: The table reports Spearman rank correlations across alternative firm-level finite single-firm welfare gains for the firm-level ranking sample over 2002–2014. $\Delta TWC_k^{\text{Baseline}}$ corresponds to the baseline network economy. $\Delta TWC_k^{\text{Economy}}$, $\Delta TWC_k^{\text{Industry}}$, and $\Delta TWC_k^{\text{Firm}}$ correspond to the economy-level, industry-level, and firm-level Ramsey benchmark reforms, respectively.

B Calibration and Measurement

This appendix documents how the empirical objects used in the welfare calculations are constructed. The main inputs are firm-level markups, firm-to-firm expenditure shares, final-demand exposure, Domar weights, and substitution elasticities. All objects are constructed separately by year.

B.1 Notation and Approximation Conventions

Rows of the input–output matrix index buyers and columns index sellers. Thus Ω_{ij} is buyer i's expenditure on seller j, divided by buyer i's sales, and $\Omega_{j\cdot}$ denotes buyer j's row. The associated revenue-based Leontief inverse is $\Psi = (I-\Omega)^{-1}$, with $\Psi_{\cdot k}$ denoting column k. Revenue-based Domar weights are denoted by λ, while cost-based objects carry tildes: $\widetilde{\Omega}$, $\widetilde{\Psi}$, and $\widetilde{\lambda}$.

The baseline wedge vector is $\Delta \log \mu = (\log \mu_1, \ldots, \log \mu_N)'$, corresponding to the marginal-cost benchmark. In Ramsey exercises, the same formulas are applied to excess wedges $\Delta \log \mu^{ex} = \log \mu - \log \mu^{\star}$ relative to a Ramsey reference markup vector $\mu^{\star}$.

All quadratic welfare objects are evaluated at the observed allocation. In particular, shares, Domar weights, elasticities, Leontief inverses, and the curvature matrix G are held fixed when computing second-order welfare losses, marginal effects, and finite single-firm reforms. We use $\mathscr{L}(\mu)$ for the log welfare loss and TWC for its empirical quadratic approximation.

B.2 Preferences and Technologies

The economy consists of a finite set of firms indexed by $i = 1, \ldots, N$ and a representative household.

Final demand. Final demand aggregates firm varieties through a CES aggregator with elasticity of substitution $\theta_0 > 0$. Let b_i denote firm i's share of final expenditure.

Production. Each firm combines intermediate inputs and primary factors through a CES production function. The elasticity of substitution across suppliers for buyer j is denoted by $\theta_j > 0$. In the baseline specification, elasticities are assumed constant across buyers:

$$\theta_j = 0.5 \quad \text{for all firms } j, \qquad \theta_0 = 0.9,$$

following the benchmark calibration in Baqaee and Farhi (2020).

B.3 Markup Measurement

Firm-level markups are measured using the production approach described in Section 3.4. For firm i in year t, the markup is

$$\mu_{it} = \frac{\beta_{s(i)}^{COGS}}{\alpha_{it}^{COGS}},$$

where α_{it}^{COGS} is the COGS share in revenue and $\beta_{s(i)}^{COGS}$ is the sector-level output elasticity assigned to firm i's sector.

To limit the influence of accounting irregularities and extreme cost-share realizations, we trim the top 1 percent and bottom 1 percent of the markup distribution within each year.

The welfare formulas use log-markup wedges. For a given year, let

$$\Delta \log \mu \equiv (\log \mu_1, \ldots, \log \mu_N)'.$$

These wedges are the primitive distortions in the baseline marginal-cost benchmark. In the Ramsey exercises, the same formulas are applied to excess wedges relative to the Ramsey benchmark.

B.4 Production Network and Final Demand

The production network is represented by the revenue-based input–output matrix Ω. Rows index buyers and columns index sellers. For buyer i and seller j,

$$\Omega_{ij} = \frac{\text{purchases by buyer } i \text{ from seller } j}{\text{sales of buyer } i}.$$

Thus, each row records the share of a buyer's sales spent on each domestic supplier. The associated revenue-based Leontief inverse is

$$\Psi \equiv (I - \Omega)^{-1}.$$

The transaction data record domestic firm-to-firm links when the annual value of transactions between a seller and a buyer is at least EUR 250. Final-demand exposure is measured as the share of firm sales absorbed outside the observed domestic B2B network. This includes sales to households, exports, and other sales not recorded as domestic firm-to-firm purchases. Let b_i denote firm i's final-demand share.

Markup wedges propagate through the production network according to

$$t \equiv \Psi \Delta \log \mu,$$

where t_i denotes the effective wedge affecting firm i after direct and indirect propagation.

B.5 Domar Weights and Network Exposure

Firm-level Domar weights measure sales relative to aggregate final expenditure:

$$\lambda_i = \frac{\text{Sales}_i}{\text{GDP}}.$$

Because firms sell intermediate inputs to one another, Domar weights need not sum to one. Their sum can exceed one in economies with intermediate production.

In the network representation, revenue-based Domar weights satisfy

$$\lambda' = b'\Psi.$$

This identity provides a consistency check between final-demand exposure, production linkages, and macroeconomic exposure. Domar weights are evaluated at the observed allocation and treated as fixed when computing second-order welfare losses and marginal effects.

B.6 Cost-Based Network Objects

The marginal welfare formulas also use cost-based network objects. Following Baqaee and Farhi (2020), define

$$\widetilde{\Omega} = \text{diag}(\mu)\,\Omega, \qquad \widetilde{\Psi} = (I - \widetilde{\Omega})^{-1}, \qquad \widetilde{\lambda}' = b'\widetilde{\Psi}.$$

These objects convert revenue-based input shares into the cost-based exposure terms that enter the closed-form expression for marginal welfare effects.

B.7 Calibration Parameters

Table 12: Calibration Parameters

Parameter	Description	Value / Source
θ_0	Elasticity of substitution in final demand	0.9
θ_j	Elasticity of substitution across intermediate inputs	0.5
μ_i	Firm-level markup	Estimated from data

Notes: Elasticity values correspond to the benchmark calibration in Baqaee and Farhi (2020), based on mid-point estimates reported in Atkeson and Burstein (2008). Firm-level markups are estimated from accounting data as described in Section 3.4.

B.8 Computed Welfare Objects

Given markups, network objects, Domar weights, and elasticities, aggregate TWC is computed from the quadratic representation derived in Appendix C. The firm-level marginal welfare effect is the gradient of this quadratic approximation,

$$M_k = (G\Delta\log\mu)_k,$$

where G is the curvature matrix of the welfare loss. Finite single-firm reform gains are computed using the same quadratic representation:

$$\Delta TWC_k = \Delta\log\mu_k \cdot M_k - \frac{1}{2}(\Delta\log\mu_k)^2 G_{kk}.$$

The aggregate calculations use the full analysis sample. Firm-level marginal and finite welfare objects are computed for the largest 10,000 firms by sales, as described in Section 3.2.

C Aggregate Welfare Losses

This appendix derives the quadratic representation used to compute aggregate TWC, starting from the buyer-level variance formula.

C.1 Buyer-Level Representation

Let $\Delta \log \mu$ denote the vector of firm-level log-markup wedges. Effective wedges propagate through the production network according to

$$t \equiv \Psi \Delta \log \mu,$$

where $\Psi = (I - \Omega)^{-1}$ is the revenue-based Leontief inverse.

The second-order welfare approximation is

$$\mathscr{L}(\mu) \approx \frac{1}{2} \sum_j \lambda_j \theta_j \operatorname{Var}_{\Omega(j)}(t), \tag{30}$$

with

$$\operatorname{Var}_{\Omega(j)}(x) = \sum_i \Omega_{ji} \left(x_i - \sum_\ell \Omega_{j\ell} x_\ell \right)^2.$$

C.2 Quadratic Form

For each buyer j, define the centered vector

$$\tilde{t}^{(j)} \equiv t - \mathbf{1} \sum_\ell \Omega_{j\ell} t_\ell,$$

so that

$$\operatorname{Var}_{\Omega(j)}(t) = \sum_i \Omega_{ji} \left(\tilde{t}_i^{(j)} \right)^2.$$

This expression is quadratic in t and can be written compactly as

$$\operatorname{Var}_{\Omega(j)}(t) = t' H_j t,$$

where H_j is the buyer-level curvature operator implied by $\Omega_{j\cdot}$.

Summing across buyers yields

$$\mathscr{L}(\mu) = \frac{1}{2} t' \left(\sum_j \lambda_j \theta_j H_j \right) t.$$

Define

$$H \equiv \sum_j \lambda_j \theta_j H_j.$$

C.3 Curvature in Markup Space

Using $t = \Psi \Delta \log \mu$, welfare losses admit the quadratic representation

$$\mathscr{L}(\mu) = \frac{1}{2}(\Delta \log \mu)' \underbrace{\Psi' H \Psi}_{G} (\Delta \log \mu). \tag{31}$$

The matrix

$$G \equiv \Psi' H \Psi$$

is symmetric and positive semi-definite.

Diagonal elements G_{kk} measure curvature with respect to firm k's markup, while off-diagonal elements capture network interactions across firms.

Equation (31) establishes that the quadratic approximation to aggregate welfare losses is fully characterized by the curvature matrix G.

D Firm-Level Welfare Effects

This appendix derives the marginal welfare effect reported in Section 2.3. The derivation follows Baqaee and Farhi (2020) and is specialized to a one-factor economy. All derivatives are evaluated at the observed distorted equilibrium. Consistent with the second-order welfare approximation used in Appendix C, network objects (Ω, Ψ, λ, $\widetilde{\lambda}$) are treated as fixed at that evaluation point.

D.1 Definition

Let $Y(\mu)$ denote equilibrium aggregate output and Y^* the efficient benchmark. Define welfare losses as

$$\mathscr{L}(\mu) \equiv \log\left(\frac{Y^*}{Y(\mu)}\right).$$

The marginal welfare effect of a change in firm k's markup wedge is [1]

[1] The derivative in (32) is a first-order general-equilibrium derivative. A marginal change in μ_k affects equilibrium prices, quantities, and hence sales shares, input shares, and network objects. Consistent with the second-order welfare approximation, all reallocations induced by the price change are captured at first order through CES share-response identities and endogenous factor-price adjustments, while higher-order terms, such as those arising from endogenous changes in Domar weights, input shares, or the Leontief inverse, are neglected.

$$M_k \equiv \frac{\partial \mathscr{L}(\mu)}{\partial \log \mu_k} = -\frac{\partial \log Y}{\partial \log \mu_k}. \tag{32}$$

The index k refers to the *selling firm* whose markup is perturbed.[2] Buyer indices j enter below because the induced price changes propagate through downstream buyers' input bundles.

D.2 Output Decomposition

Assume a one-factor economy with wage w and define the general-equilibrium wage response

$$s_k \equiv \frac{\partial \log w}{\partial \log \mu_k}.$$

Lemma D.1 (Output elasticity decomposition)**.** *In a one-factor economy,*

$$\frac{\partial \log Y}{\partial \log \mu_k} = -\widetilde{\lambda}_k - s_k, \tag{33}$$

where $\widetilde{\lambda}_k$ is the cost-based Domar weight.

Proof. The perturbation $d \log \mu_k$ creates a wedge between firm k's price and marginal cost. Holding the wage fixed, the first-order effect on $\log Y$ is given by the cost-share (envelope) of that wedge, which equals $-\widetilde{\lambda}_k \, d \log \mu_k$ (this is the envelope step in Baqaee and Farhi (2020) and is the definition of the cost-based Domar weight as the derivative of $\log Y$ with respect to a firm-level wedge, holding factor prices fixed).

In general equilibrium, the wage also adjusts. Since labor is the only primary factor, the wage response affects all firms' unit costs proportionally to their value-added content. The induced first-order change in $\log Y$ from this common factor-price movement is $-s_k \, d \log \mu_k$. Summing both first-order channels yields (33). □

Combining (32) with Lemma D.1 implies the fundamental identity

$$M_k = \widetilde{\lambda}_k + s_k. \tag{34}$$

Thus, the problem reduces to solving for s_k.

[2]The index k refers to the *selling firm* whose markup μ_k is marginally perturbed. Buyer indices j appear in subsequent expressions because a change in a seller's markup propagates through the production network and induces reallocation within downstream buyers' input bundles. All variance and covariance terms are therefore evaluated at the buyer level, while the distortion itself originates at the seller node k.

D.3 Propagation of unit-cost changes

Let Ω denote the revenue-based input–output matrix and

$$\Psi \equiv (I - \Omega)^{-1}$$

the associated Leontief inverse. Let $\Psi_{\cdot k}$ be column k and $\Psi_{\cdot L}$ the value-added column.

A marginal change in $\log \mu_k$ affects firm i's unit cost through direct propagation and the wage response:

$$\frac{\partial \log c_i}{\partial \log \mu_k} = \Psi_{ik} + s_k \Psi_{iL}. \tag{35}$$

D.4 CES share responses and covariance identity

Consider buyer j with CES aggregation across its inputs and elasticity of substitution θ_j. Let Ω_{ji} denote the revenue share spent by buyer j on supplier i. The CES demand system implies

$$d \log \Omega_{ji} = (\theta_j - 1) \left(d \log p_i - \sum_{\ell} \Omega_{j\ell} d \log p_\ell \right). \tag{36}$$

Define buyer-level moments under row weights $\Omega_{j\cdot}$:

$$\mathbb{E}_{\Omega(j)}[x] = \sum_i \Omega_{ji} x_i, \quad \mathrm{Var}_{\Omega(j)}(x) = \mathbb{E}_{\Omega(j)}[x^2] - \mathbb{E}_{\Omega(j)}[x]^2,$$

$$\mathrm{Cov}_{\Omega(j)}(x,y) = \mathbb{E}_{\Omega(j)}[xy] - \mathbb{E}_{\Omega(j)}[x]\mathbb{E}_{\Omega(j)}[y].$$

Lemma D.2 (Covariance identity)**.** *For any vector x,*

$$\sum_i x_i \, d\Omega_{ji} = (\theta_j - 1) \, \mathrm{Cov}_{\Omega(j)}\big(x, d \log p\big). \tag{37}$$

Proof. From (36), $d\Omega_{ji} = \Omega_{ji} \, d \log \Omega_{ji}$ and hence

$$\sum_i x_i d\Omega_{ji} = (\theta_j - 1) \sum_i x_i \Omega_{ji} \left(d \log p_i - \sum_{\ell} \Omega_{j\ell} d \log p_\ell \right),$$

which equals $(\theta_j - 1) \left(\mathbb{E}_{\Omega(j)}[x \, d \log p] - \mathbb{E}_{\Omega(j)}[x] \mathbb{E}_{\Omega(j)}[d \log p] \right)$, i.e. (37). □

D.5 Wage Normalization and Closed Form for s_k

Substituting (35) into Lemma D.2 and imposing value-added market clearing (labor market clearing) yields the linear wage equation

$$s_k = -\lambda_k \Psi_{kL} - \sum_j \lambda_j(\theta_j - 1)\,\mathrm{Cov}_{\Omega(j)}(\Psi_{\cdot L}, \Psi_{\cdot k}) - s_k \sum_j \lambda_j(\theta_j - 1)\,\mathrm{Var}_{\Omega(j)}(\Psi_{\cdot L}). \tag{38}$$

Define the scalars

$$A \equiv \sum_j \lambda_j(\theta_j - 1)\ \mathrm{Var}_{\Omega(j)}(\Psi_{\cdot L}), \tag{39}$$

$$B_k \equiv \sum_j \lambda_j(\theta_j - 1)\ \mathrm{Cov}_{\Omega(j)}(\Psi_{\cdot L}, \Psi_{\cdot k}). \tag{40}$$

Then (38) becomes $s_k = -\lambda_k \Psi_{kL} - B_k - A s_k$, so

$$s_k = \frac{-\lambda_k \Psi_{kL} - B_k}{1 + A}. \tag{41}$$

Substituting (41) into (34) yields the closed form

$$M_k = \widetilde{\lambda}_k + \frac{-\lambda_k \Psi_{kL} - B_k}{1 + A}. \tag{42}$$

D.6 Gradient Representation of Marginal Effects

Appendix C shows that the empirical TWC approximation can be written as

$$\mathrm{TWC}(\Delta \log \mu) = \frac{1}{2} (\Delta \log \mu)^\top G (\Delta \log \mu), \tag{43}$$

for a symmetric positive semi-definite matrix G.

Differentiating (43) gives

$$\frac{\partial\, \mathrm{TWC}}{\partial\, \Delta \log \mu_k} = \big(G \Delta \log \mu\big)_k. \tag{44}$$

Under the approximation convention above, the marginal welfare effect used in the empirical rankings equals this gradient:

$$M_k = \frac{\partial\, \mathrm{TWC}}{\partial\, \Delta \log \mu_k} = \big(G \Delta \log \mu\big)_k. \tag{45}$$

Equation (45) is the precise sense in which M_k measures local welfare leverage: for a small change $d\Delta\log\mu_k$ holding other markup wedges fixed,

$$d\,\mathrm{TWC} \approx M_k\, d\Delta\log\mu_k.$$

D.7 Finite Single-Firm Reforms

Consider a reform that sets $\Delta\log\mu_k$ to the benchmark value 0 while keeping all other wedges fixed:

$$\Delta\log\mu_k' = 0, \qquad \Delta\log\mu_i' = \Delta\log\mu_i \quad \text{for } i \neq k.$$

Define the finite single-firm welfare gain

$$\Delta TWC_k \equiv \mathrm{TWC}(\Delta\log\mu) - \mathrm{TWC}(\Delta\log\mu').$$

Lemma D.3. *The finite single-firm welfare gain satisfies*

$$\Delta TWC_k = \Delta\log\mu_k \cdot M_k - \frac{1}{2}\,(\Delta\log\mu_k)^2\, G_{kk}. \tag{46}$$

Proof. Using (43),

$$\mathrm{TWC}(\Delta\log\mu) = \frac{1}{2}\sum_{i,j} G_{ij}\Delta\log\mu_i\Delta\log\mu_j.$$

Setting $\Delta\log\mu_k = 0$ removes all terms in the sum containing index k. Hence the difference equals

$$\sum_j G_{kj}\Delta\log\mu_k\Delta\log\mu_j - \frac{1}{2}\,G_{kk}(\Delta\log\mu_k)^2.$$

The first term is $\Delta\log\mu_k\,(G\Delta\log\mu)_k = \Delta\log\mu_k\, M_k$ by (45), which yields (46). □

Diagonal curvature. It is useful to name the curvature term:

$$C_k \;\equiv\; G_{kk}. \tag{47}$$

Note that C_k is *not* the same object as B_k in (40): B_k is a covariance term entering the wage response, whereas $C_k = G_{kk}$ is the diagonal curvature of the quadratic welfare loss. Using a distinct symbol avoids overloading notation.

Using the variance representation of G from Appendix C, the diagonal element has the explicit form

$$G_{kk} = \sum_j \lambda_j \theta_j \operatorname{Var}_{\Omega(j)}\left(\Psi_{\cdot k}\right). \tag{48}$$

Equations (46)–(48) show that finite single-firm welfare gains equal the integrated marginal effect minus a second-order curvature correction.

E Interpretation and Sign of Marginal Welfare Effects

This appendix records two interpretation points for the marginal welfare effect M_k used in the empirical rankings. First, M_k is a local derivative, not a stand-alone contribution to aggregate losses. Second, its sign can be ambiguous in a second-best environment with multiple markup wedges.

Using the quadratic representation,

$$\mathscr{L}(\mu) = \frac{1}{2}(\Delta \log \mu)' G (\Delta \log \mu),$$

the marginal welfare effect is

$$M_k = (G \Delta \log \mu)_k.$$

It is therefore the slope of aggregate welfare losses with respect to firm k's markup wedge, holding other wedges fixed. For a small perturbation,

$$d\mathscr{L} \approx M_k \, d\Delta \log \mu_k.$$

The sign of M_k reflects both direct exposure and general-equilibrium reallocation. From Appendix D,

$$M_k = \widetilde{\lambda}_k + \frac{-\lambda_k \Psi_{kL} - B_k}{1 + A}.$$

The first term is positive and captures firm k's direct exposure. The second term summarizes wage normalization and input-substitution effects. It can be positive or negative depending on the existing configuration of distortions and the firm's network position. A negative value of M_k therefore does not mean that markups are welfare-improving; it means that a marginal reduction in one wedge can locally increase measured welfare losses when other wedges remain in place. This is the usual second-best logic of Lipsey and Lancaster (1956).

Marginal effects also do not add up to aggregate TWC. In particular,

$$\sum_k M_k \Delta \log \mu_k = (\Delta \log \mu)' G (\Delta \log \mu) = 2\mathscr{L}(\mu).$$

The aggregate welfare loss is a second-order object, while M_k is a first-order derivative evaluated at the observed distorted allocation. This is why the empirical analysis reports both marginal welfare effects and finite single-firm welfare gains.

F Horizontal Economy Benchmark

This appendix constructs a horizontal benchmark that shuts down production-network propagation while keeping substitution elasticities, observed markups, and final-demand shares fixed. The purpose is to isolate the role of input–output linkages in generating aggregate welfare losses.

F.1 Definition of the Horizontal Economy

Let Ω_{FF} denote the firm-to-firm block of the revenue-based input–output matrix. The horizontal benchmark imposes

$$\Omega_{FF} = 0,$$

so that firms do not purchase intermediate inputs from other firms. The corresponding Leontief inverse over firms satisfies

$$\Psi_{FF} = (I - \Omega_{FF})^{-1} = I.$$

All other primitives such as substitution elasticities, final-demand shares, and observed markups are held fixed relative to the baseline network economy. The horizontal economy therefore isolates the propagation channel by restricting the production structure while preserving all other objects.

F.2 Total Welfare Cost in the Horizontal Economy

When $\Omega_{FF} = 0$, there is no upstream or downstream propagation of distortions through intermediate inputs. Effective wedges coincide with firm-level markup wedges, and within-buyer substitution across intermediate suppliers is absent.

In our one-factor environment, nominal GDP equals aggregate value added. The natural horizontal Domar weights are therefore firms' value-added shares:

$$\lambda_k^H \equiv \frac{VA_k}{\sum_i VA_i}, \qquad \sum_k \lambda_k^H = 1. \tag{49}$$

Because there is no within-buyer substitution across intermediate suppliers, welfare losses reduce to dispersion of firm-level markup wedges across final demand. The horizontal Total Welfare Cost is therefore approximated by

$$\mathscr{L}^H(\mu) \approx \frac{1}{2}\theta_0 \operatorname{Var}_{\lambda^H}\big(\Delta \log \mu\big), \tag{50}$$

where

$$\operatorname{Var}_{\lambda^H}(x) = \sum_k \lambda_k^H \big(x_k - \bar{x}_{\lambda^H}\big)^2, \qquad \bar{x}_{\lambda^H} = \sum_j \lambda_j^H x_j.$$

Hence, in the horizontal benchmark, welfare losses arise purely from cross-firm dispersion in markup wedges, scaled by value-added shares.

G Fixed Costs and Ramsey Benchmarks

This appendix incorporates fixed costs into the welfare-accounting framework and constructs second-best benchmarks. The key message is that fixed costs affect the *reference markup vector* relative to which wedges are measured, but do not change the network propagation structure governing marginal distortions and curvature. The empirical Ramsey exercises should therefore be read as robustness benchmarks: they ask whether the main results survive when part of observed markups is interpreted as necessary to finance fixed costs.

G.1 Fixed Costs and the McKenzie Representation

Consider a firm i with technology

$$y_i = f_i(x_i) - F_i,$$

where x_i denotes variable inputs and $F_i > 0$ is a fixed cost. This violates constant returns in variable inputs.

Following McKenzie (1959) (see also Baqaee and Farhi (2020)), such a technology can be represented equivalently as constant returns to scale in an augmented input vector by introducing a fixed factor K_i supplied in

fixed quantity:

$$y_i = \tilde{f}_i(x_i, K_i),$$

where $\tilde{f}_i$ is constant returns to scale in (x_i, K_i) and the supply of K_i is fixed. Payments to K_i absorb the non-convex component F_i (or, more generally, scale rents). Since K_i is fixed, this transformation preserves the first-order conditions for variable inputs and therefore leaves the propagation of marginal wedges through the production network unchanged.

In particular, holding the firm-to-firm input-share matrix Ω_{FF} fixed implies that the Leontief inverse $\Psi_{FF} = (I - \Omega_{FF})^{-1}$ and the propagated wedges $t_F = \Psi_{FF}\Delta \log \mu_F$ are unchanged. Consequently, the curvature objects entering the quadratic welfare approximation (and the marginal-effect formula) are invariant to whether fixed costs are represented explicitly or via fixed factors.

G.2 Ramsey Logic with Fixed Costs

When production involves fixed costs, marginal-cost pricing is generally infeasible because firms must generate sufficient operating surplus to finance non-convex expenditures. Efficiency must therefore be evaluated relative to a second-best (Ramsey) allocation rather than a zero-markup allocation.

Let $\mu = (\mu_i)_i$ denote firm-level markups and let $W(\mu)$ denote aggregate welfare or real output. A generic Ramsey problem is

$$\max_{\mu} W(\mu) \qquad \text{s.t.} \qquad \sum_{i \in \mathscr{G}} \Pi_i(\mu) \geq \sum_{i \in \mathscr{G}} F_i, \tag{51}$$

where $\mathscr{G}$ denotes the set of firms (or groups) over which financing is required, $\Pi_i(\mu)$ is the operating surplus generated by markups, and F_i denotes fixed costs.

Under standard markup accounting, operating surplus generated by pricing above marginal cost equals the Lerner index times sales:

$$\Pi_i(\mu) = (p_i - MC_i)q_i = \frac{\mu_i - 1}{\mu_i} R_i, \tag{52}$$

where $R_i \equiv p_i q_i$ is nominal revenue. Equation (52) measures *gross* operating surplus before fixed costs are paid. Net operating profit would therefore be $\Pi_i(\mu) - F_i$.

Following the classical public-finance approach of Ramsey (1927), Diamond and Mirrlees (1971), and Atkinson and Stiglitz (1980), the Ramsey first-order condition equates the marginal welfare cost of a higher

markup to the marginal financing benefit of the additional operating surplus:

$$Y(\mu)M_i = \eta \frac{\partial \Pi_i(\mu)}{\partial \log \mu_i}. \tag{53}$$

The fully optimal allocation would require firm-level demand elasticities and the planner multiplier η. Because these objects are not identified in the administrative data, the empirical analysis uses transparent financing benchmarks instead of estimating the full firm-specific Ramsey rule.

G.3 Empirical Measurement of Fixed Costs

To implement the Ramsey benchmark, we require an empirical measure of fixed costs. We use accounting information on operating profits to separate gross markup revenue from fixed operating expenditures.

Let R_i denote firm revenue and let $EBIT_i$ denote earnings before interest and taxes. Gross operating surplus generated by markup pricing is

$$\Pi_i = \frac{\mu_i - 1}{\mu_i} R_i.$$

Under the maintained accounting interpretation, EBIT measures operating profit after fixed operating expenditures. We therefore interpret

$$EBIT_i = \Pi_i - F_i$$

as defining fixed operating expenditures residually.

Rearranging yields an empirical measure of fixed costs:

$$F_i = \frac{\mu_i - 1}{\mu_i} R_i - EBIT_i. \tag{54}$$

This construction makes the fixed-cost measure internally consistent with observed markups, observed operating profits, and the financing constraint used to define the Ramsey benchmarks. Only the portion of markup revenue required to cover fixed costs enters the benchmark markup. Any remaining markup revenue is treated as an excess wedge in the welfare accounting.

G.4 Empirical Ramsey Benchmarks

The Ramsey problem in (51) characterizes the fully optimal second-best allocation when fixed costs must be financed through distortionary markups. In general, the first-order condition for firm i implies an inverse-elasticity rule: markups are set higher where the welfare cost of raising revenue is lower.

Implementing the full firm-level inverse-elasticity rule requires knowledge of firm-specific demand elasticities. Such elasticities are not directly identified in our administrative data. We therefore do not estimate the fully optimal Ramsey vector. Instead, we impose parsimonious reference vectors that preserve the financing logic while remaining consistent with the available information.

Economy-level benchmark. The economy-level benchmark imposes a common reference markup $\mu^\star$ on all firms. Let

$$R \equiv \sum_i R_i, \qquad F \equiv \sum_i F_i.$$

The common markup is chosen so that aggregate operating surplus finances aggregate fixed costs:

$$\frac{\mu^\star - 1}{\mu^\star} R = F.$$

Solving gives

$$\mu^\star = \frac{1}{1 - F/R},$$

provided $F < R$. This benchmark removes the common markup component needed to finance aggregate fixed costs, while leaving cross-firm differences in excess markups as distortions.

Industry-level benchmark. We assume that within each industry g, firms face a common reference markup $\mu_g^\star$. Let

$$R_g \equiv \sum_{i \in g} R_i, \qquad F_g \equiv \sum_{i \in g} F_i.$$

Under this restriction, aggregate operating surplus generated in industry g is

$$\sum_{i \in g} \Pi_i(\mu_g^\star) = \frac{\mu_g^\star - 1}{\mu_g^\star} R_g.$$

Imposing that industry-level surplus finances industry-level fixed costs yields

$$\frac{\mu_g^\star - 1}{\mu_g^\star} R_g = F_g. \tag{55}$$

Solving for the financing-consistent industry markup gives

$$\mu_g^\star = \frac{1}{1 - F_g/R_g}, \tag{56}$$

provided $F_g < R_g$.

Relation to inverse-elasticity logic. The industry-level restriction is a coarse implementation of Ramsey logic under limited elasticity information. It allows benchmark markups to differ across industries while avoiding strong assumptions about firm-level demand.

Firm-level benchmark. The firm-level benchmark allows the fixed-cost-financing markup to vary across firms. For each firm i, the benchmark markup $\mu_i^\star$ satisfies

$$\frac{\mu_i^\star - 1}{\mu_i^\star} R_i = F_i,$$

so that

$$\mu_i^\star = \frac{1}{1 - F_i/R_i},$$

provided $F_i < R_i$. This benchmark is the most flexible of the three. It interprets firm-specific fixed-cost requirements as non-distortionary benchmark markups and treats only markups above that level as excess wedges.

Role of the benchmarks. The economy-, industry-, and firm-level benchmarks differ only in the degree of heterogeneity allowed in the reference markup vector. They are not alternative estimates of a fully optimal regulatory policy. Their purpose is to test whether aggregate welfare losses and firm-level rankings depend on treating all observed markups as distortions relative to marginal-cost pricing.

G.5 Excess Wedges, Excess TWC, and Excess Marginal Effects

The Ramsey benchmarks shift the reference markup vector, but do not alter the dispersion-based structure of welfare losses. Since the Baqaee–Farhi quadratic formula depends on dispersion of effective wedges and on

network propagation, changing the benchmark from marginal-cost pricing to $\mu^\star$ does not alter the curvature matrix G or the propagation objects (Ω, Ψ, λ).

Given the Ramsey benchmark, define excess log-markup wedges as

$$\Delta \log \mu_i^{ex} \equiv \log \mu_i - \log \mu_i^\star.$$

Conceptual definition. The excess Total Welfare Cost is the welfare gap between the Ramsey benchmark and the observed economy:

$$\text{TWC}^{ex} \equiv W(\mu^\star) - W(\mu). \tag{57}$$

Operational computation. We compute TWC^{ex} using the same Baqaee–Farhi quadratic approximation as in the baseline analysis, evaluated at the excess wedge vector. In particular, all network objects (Ω, Ψ, λ) and curvature parameters are unchanged, and only the reference point for markup wedges changes:

$$\text{TWC}^{ex} \approx \text{TWC}(\Delta \log \mu^{ex}).$$

Excess marginal effects. Excess marginal welfare effects are defined as derivatives with respect to excess wedges:

$$M_k^{ex} \equiv \frac{\partial \, \text{TWC}^{ex}}{\partial \, \Delta \log \mu_k^{ex}}.$$

Since the benchmark $W(\mu^\star)$ is fixed, M_k^{ex} is computed using the same curvature matrix as in the baseline analysis, evaluated at the excess wedge vector. The interpretation is shifted from "above marginal cost" to "above Ramsey":

$$M_k^{ex} = (G \Delta \log \mu^{ex})_k.$$

Excess finite single-firm welfare gains. The corresponding finite single-firm gain from setting firm k's excess wedge to zero is

$$\Delta TWC_k^{ex} = \Delta \log \mu_k^{ex} \cdot M_k^{ex} - \frac{1}{2} (\Delta \log \mu_k^{ex})^2 G_{kk}.$$

These are the firm-level objects used to compare rankings across the baseline and Ramsey benchmarks.

G.6 Scope

The welfare objects in this paper measure allocative inefficiency conditional on the observed set of firms. Welfare effects associated with endogenous entry, exit, or product variety are outside the scope of the framework.